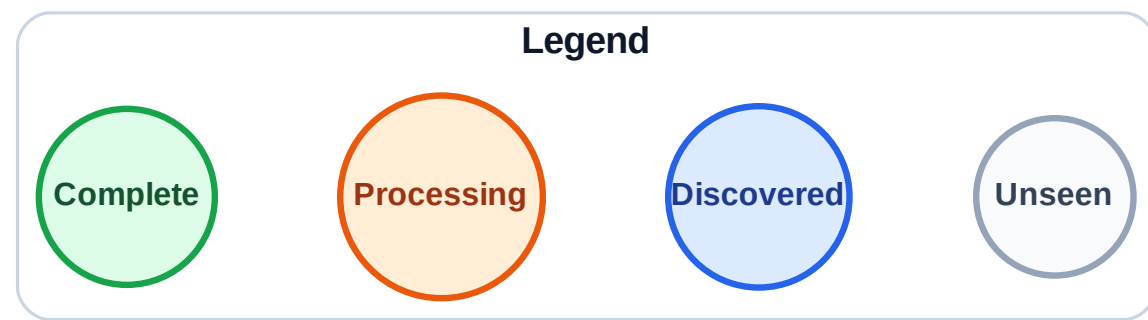
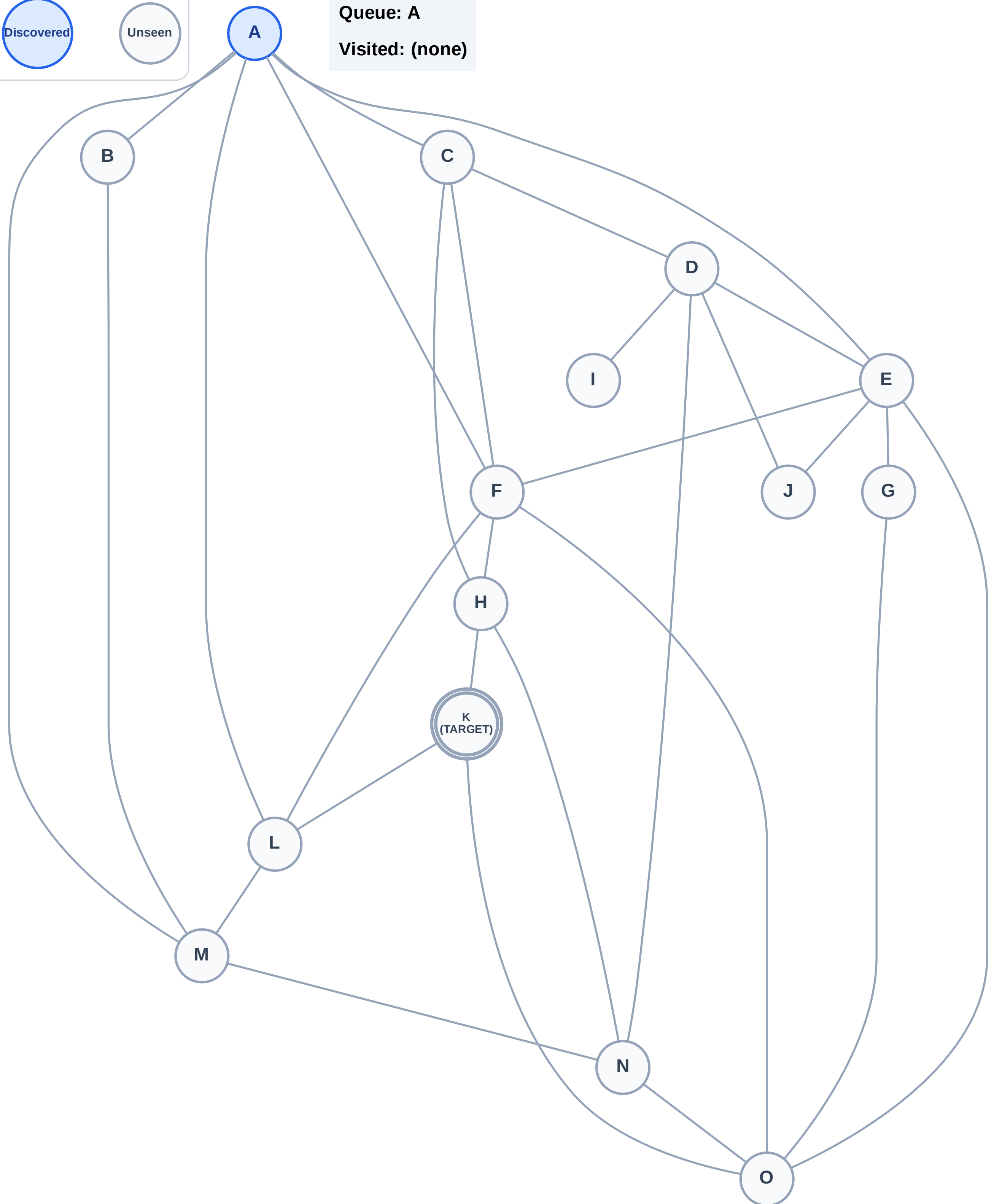


BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

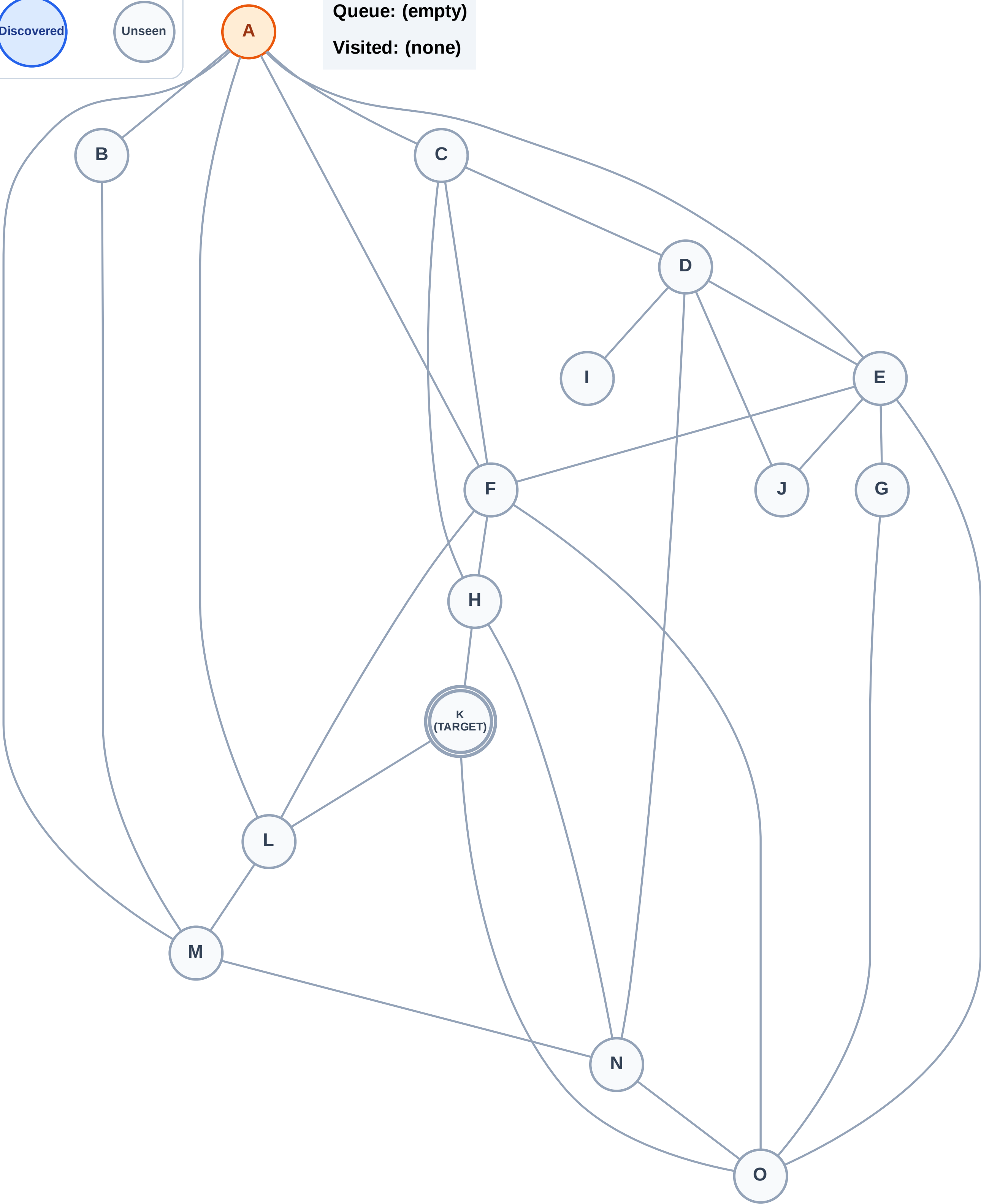
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



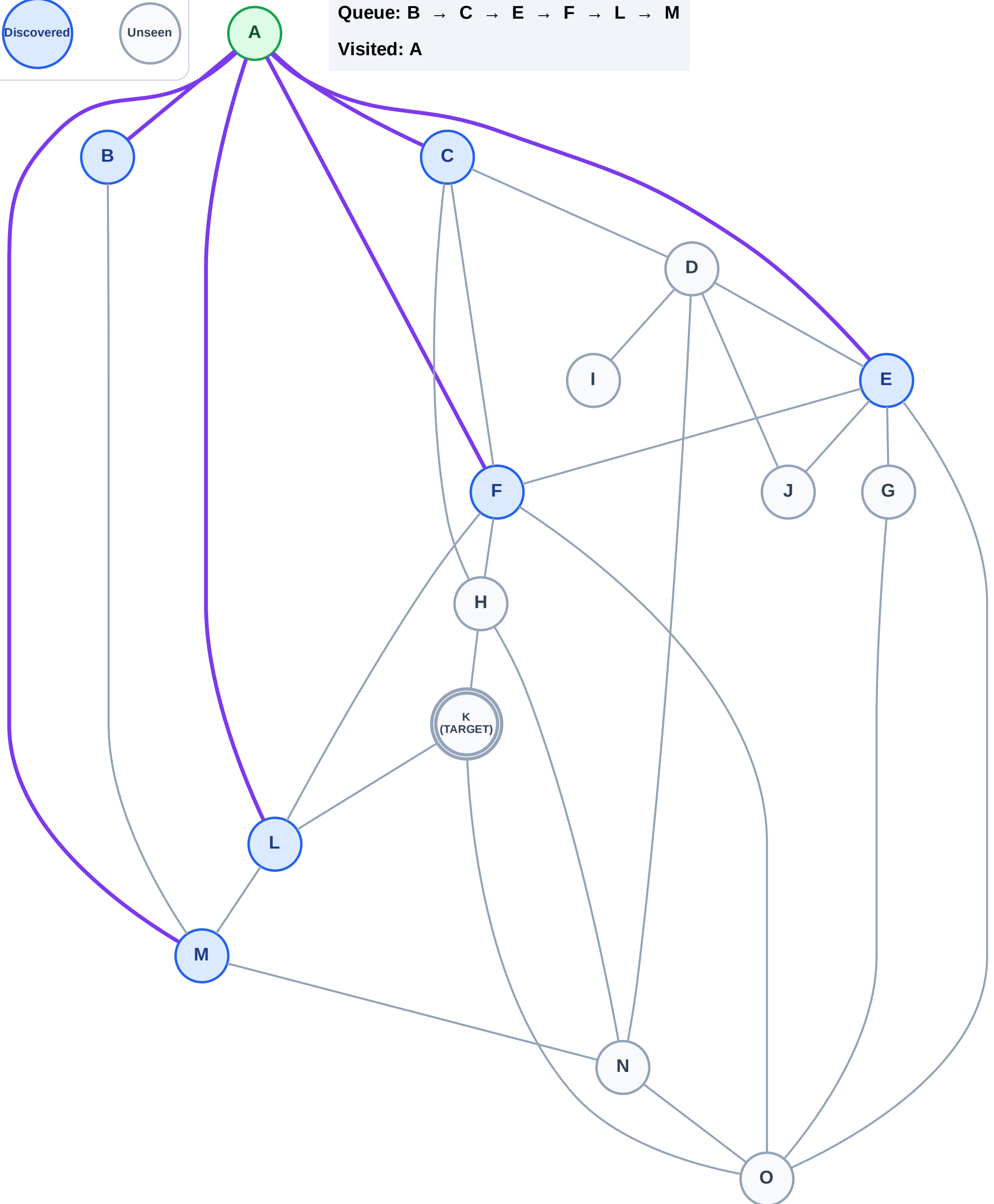
BFS Traversal

Step 3: Discover B, C, E, F, L, M from A

Legend



Queue: B → C → E → F → L → M
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

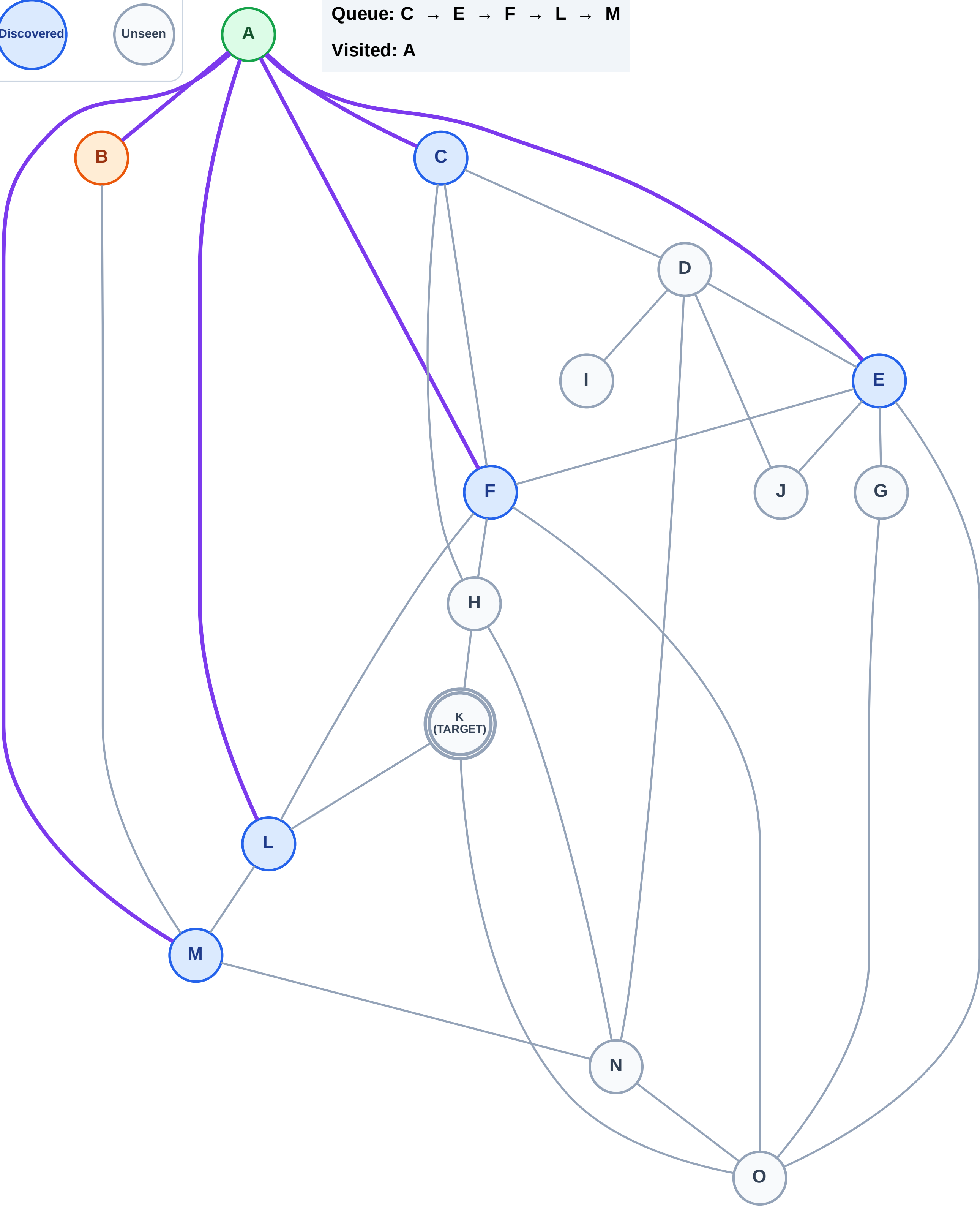
Processing

Discovered

Unseen

Queue: C → E → F → L → M

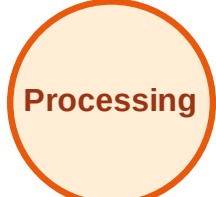
Visited: A



BFS Traversal

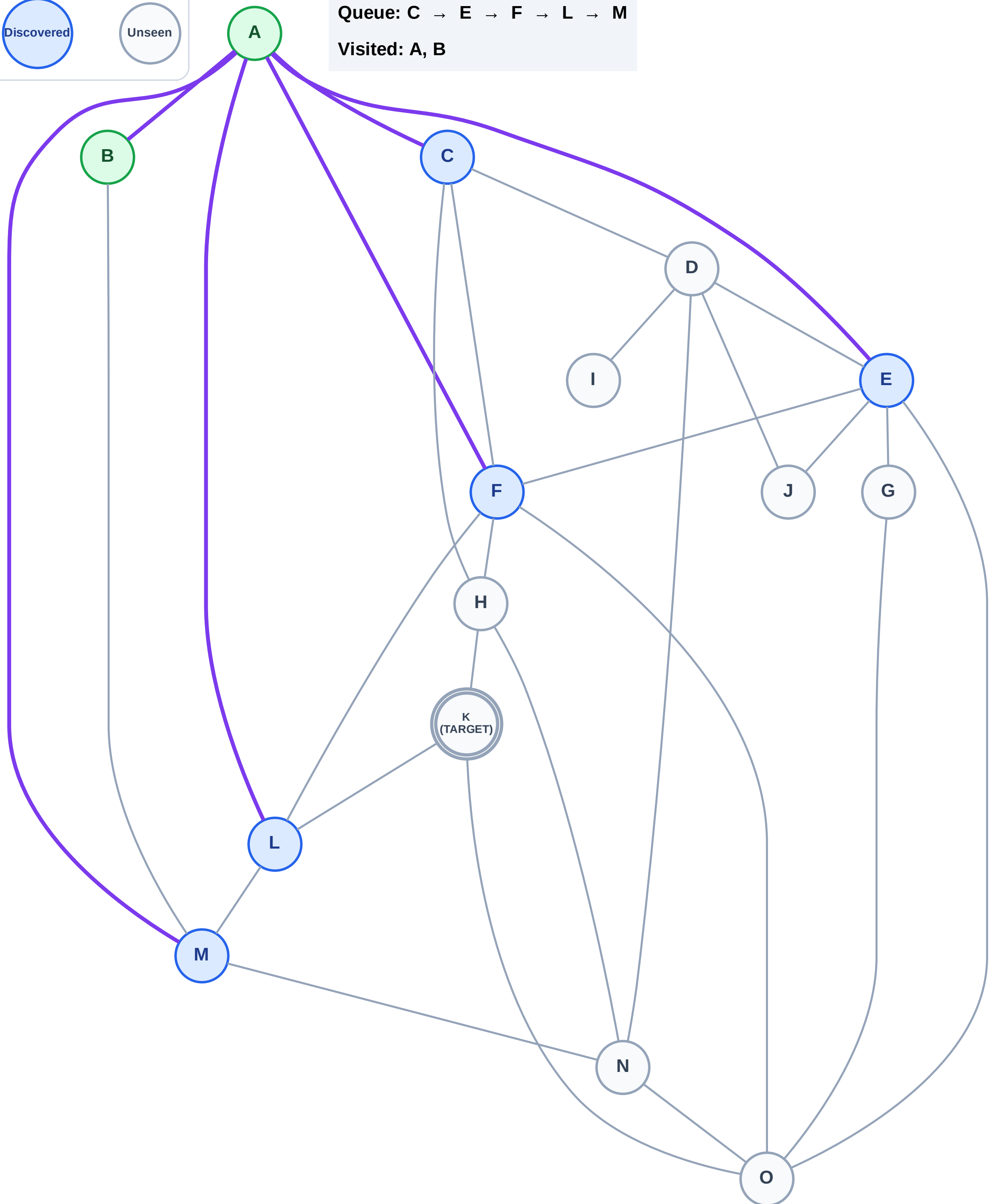
Step 5: No new neighbors from B

Legend

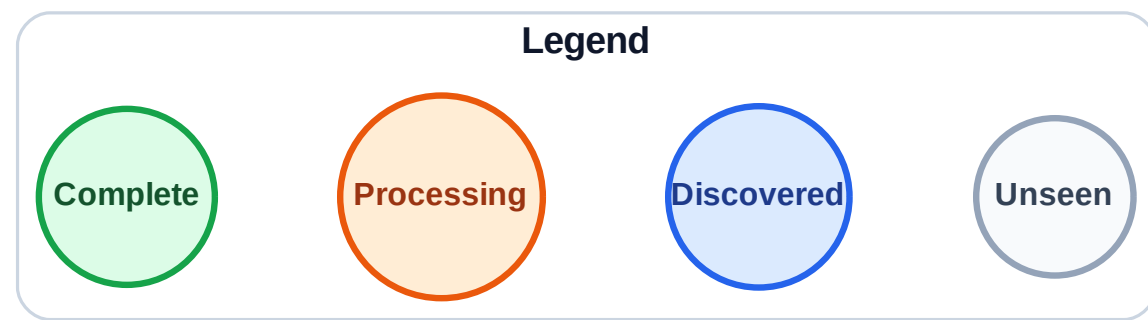


Queue: C → E → F → L → M

Visited: A, B

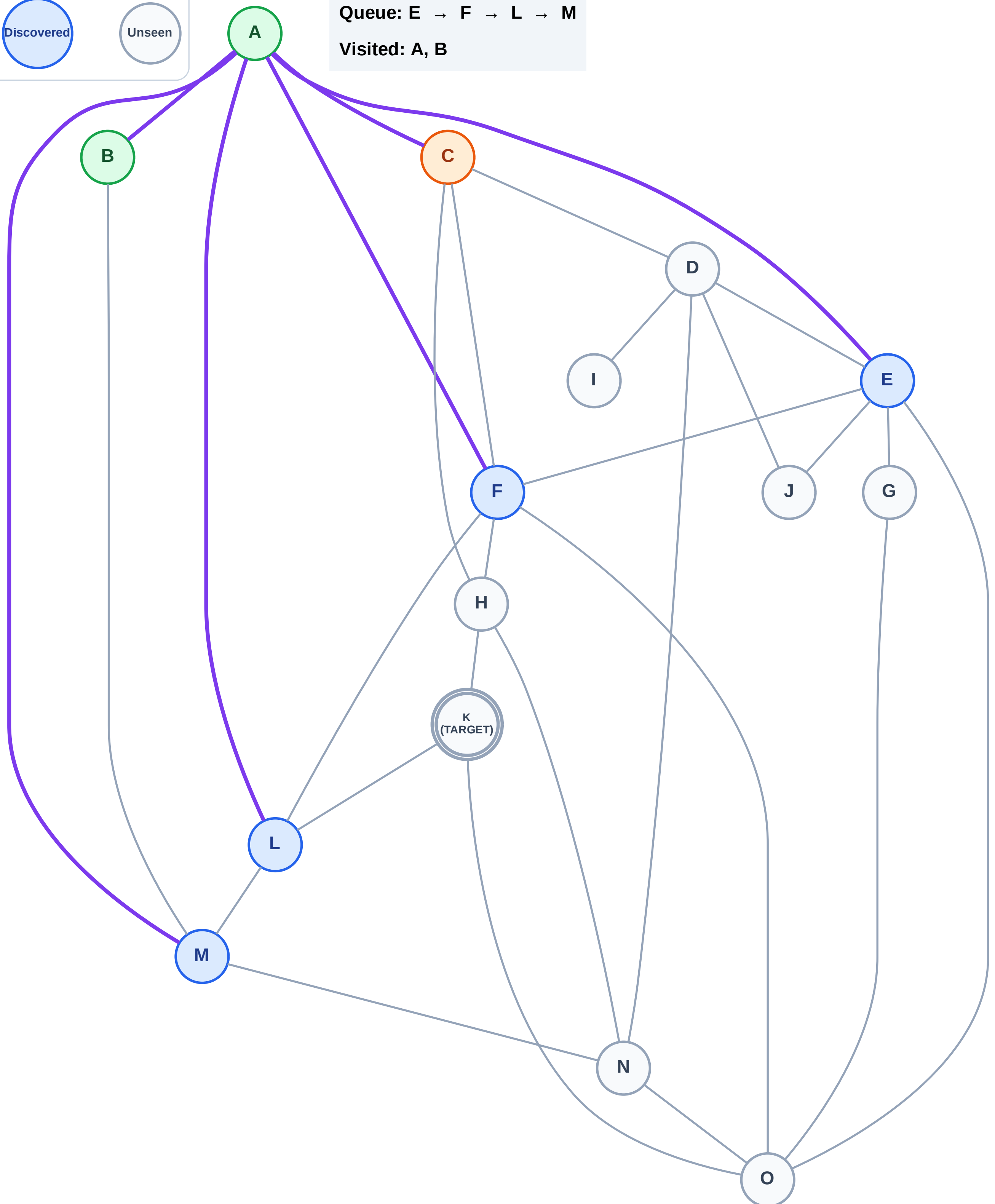


Step 6: Process node C



Queue: E → F → L → M

Visited: A, B



BFS Traversal

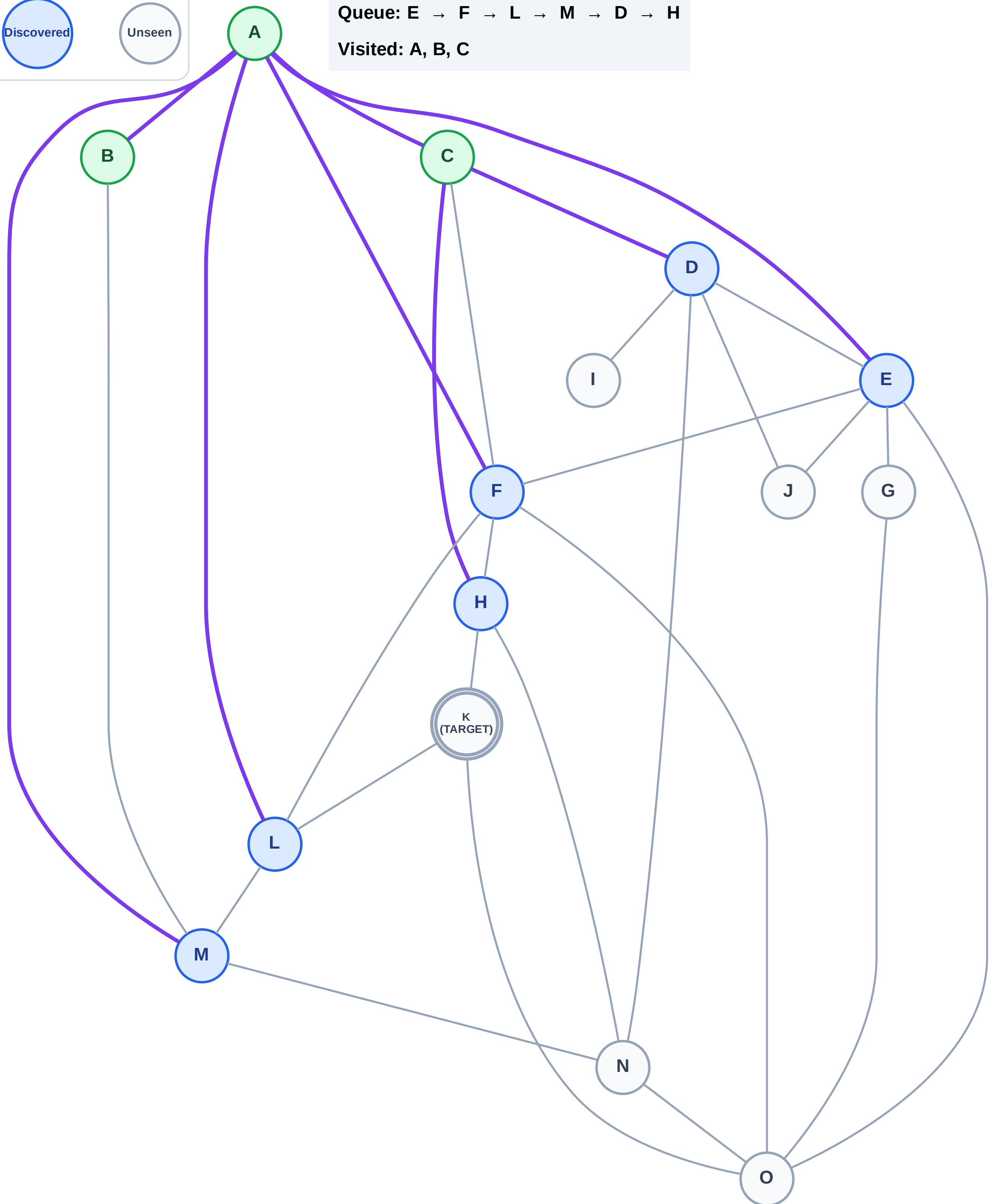
Step 7: Discover D, H from C

Legend



Queue: E → F → L → M → D → H

Visited: A, B, C



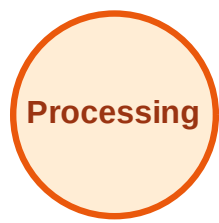
BFS Traversal

Step 8: Process node E

Legend



Complete



Processing



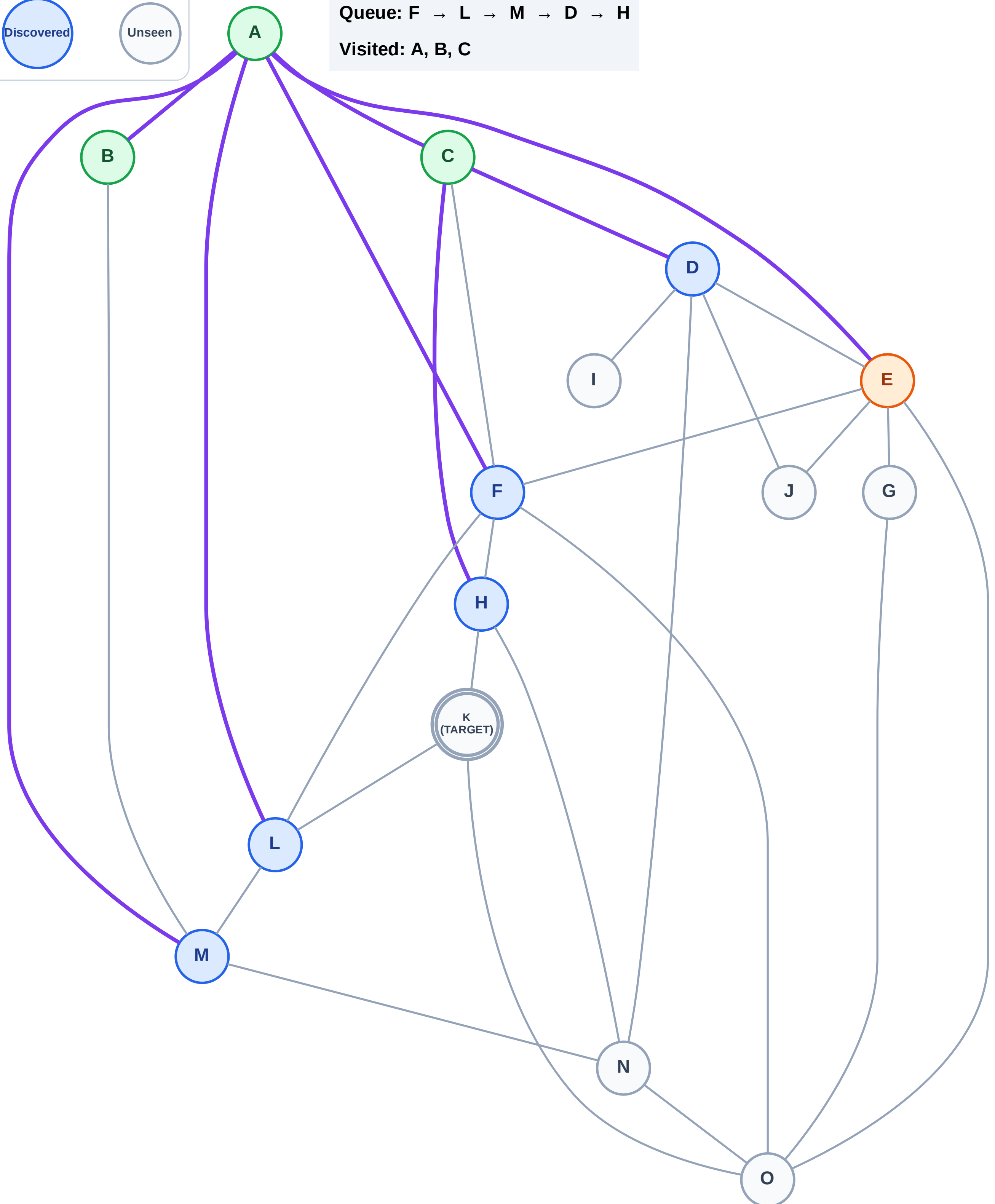
Discovered



Unseen

Queue: F → L → M → D → H

Visited: A, B, C



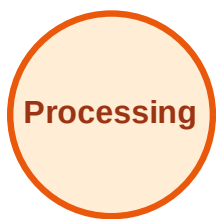
BFS Traversal

Step 9: Discover G, J, O from E

Legend



Complete



Processing



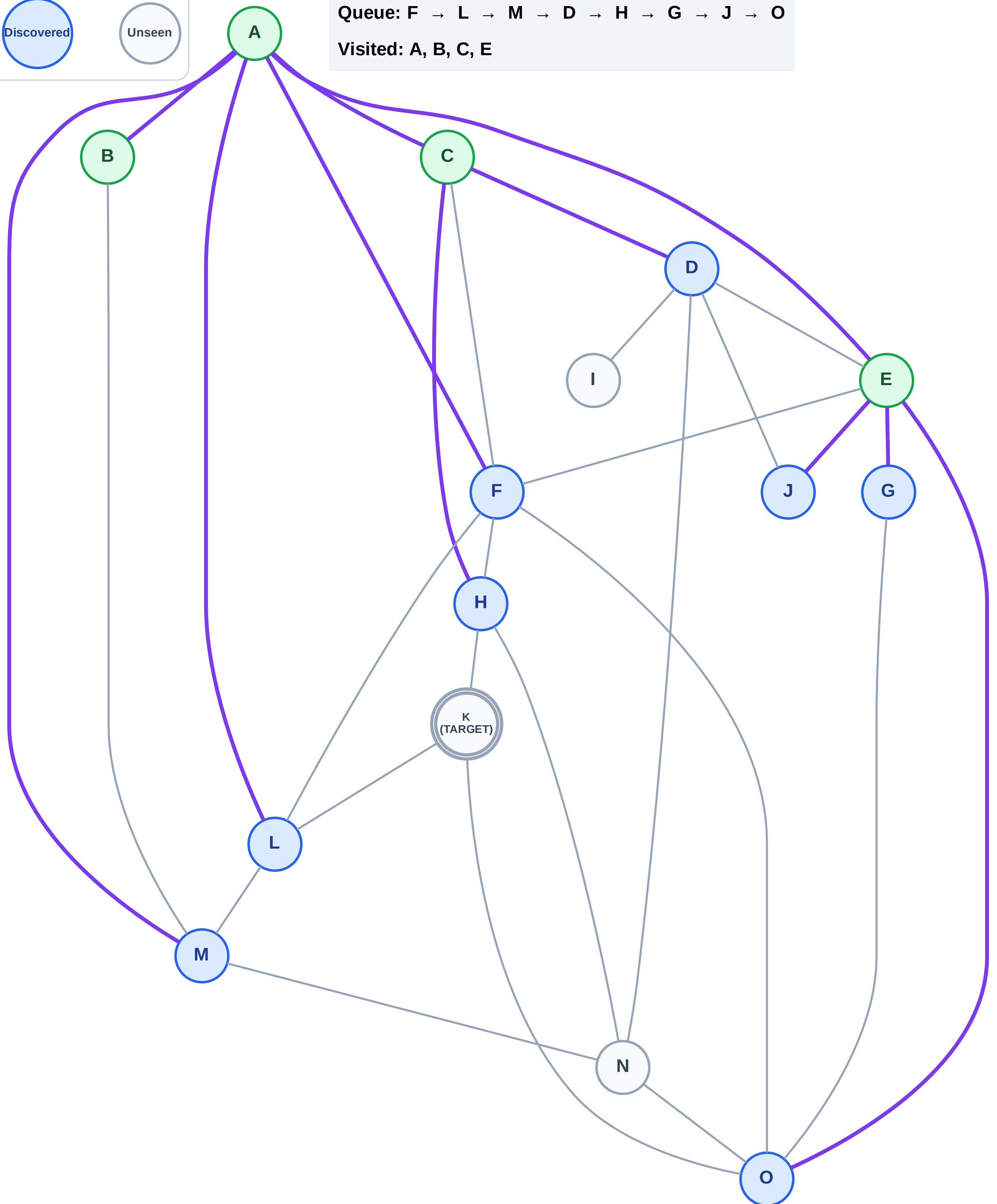
Discovered



Unseen

Queue: F → L → M → D → H → G → J → O

Visited: A, B, C, E



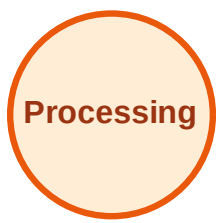
BFS Traversal

Step 10: Process node F

Legend



Complete



Processing



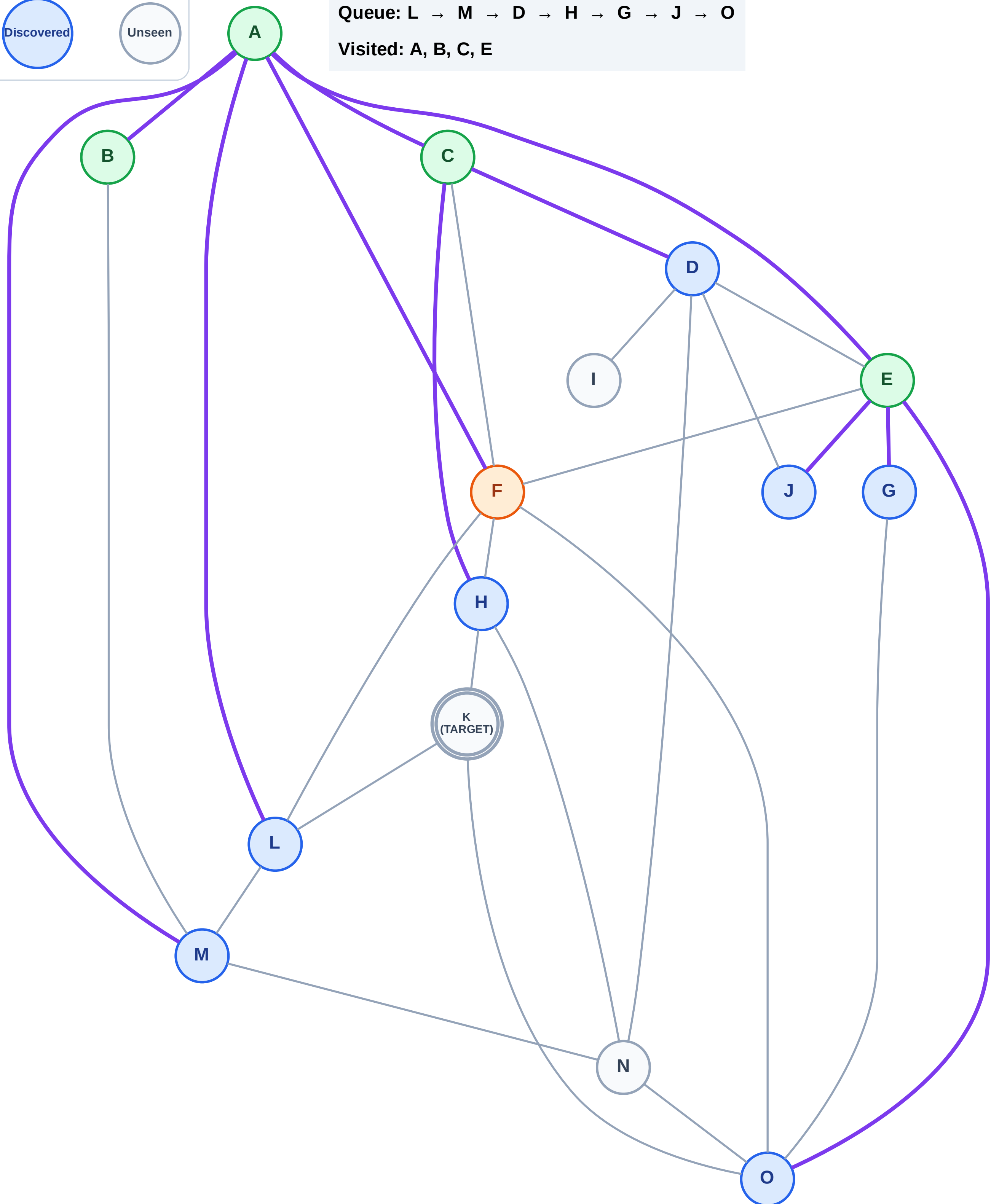
Discovered



Unseen

Queue: L → M → D → H → G → J → O

Visited: A, B, C, E



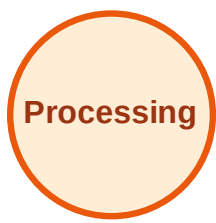
BFS Traversal

Step 11: No new neighbors from F

Legend



Complete



Processing



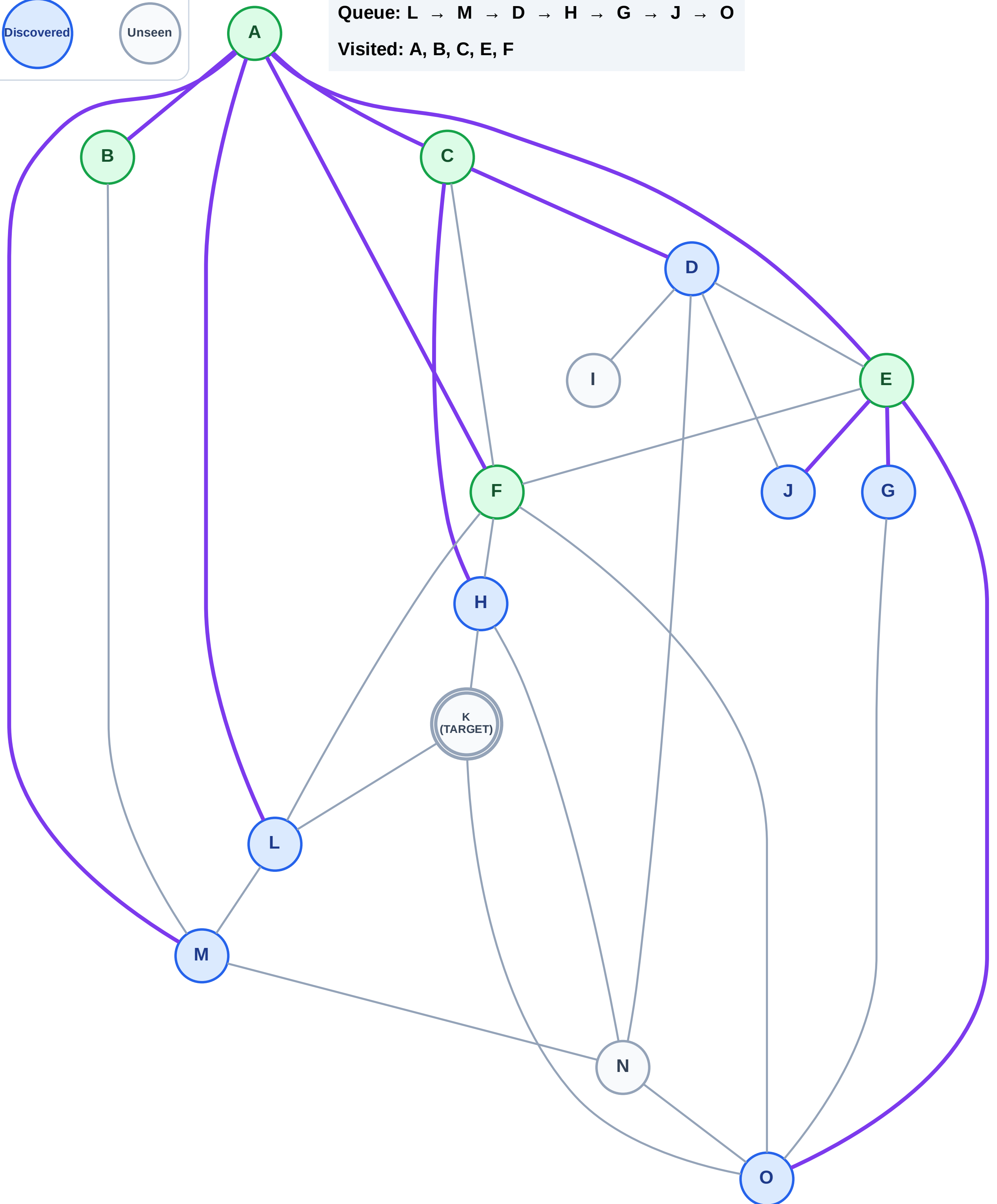
Discovered



Unseen

Queue: L → M → D → H → G → J → O

Visited: A, B, C, E, F



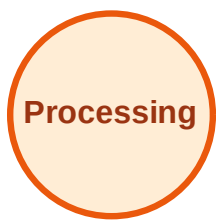
BFS Traversal

Step 12: Process node L

Legend



Complete



Processing



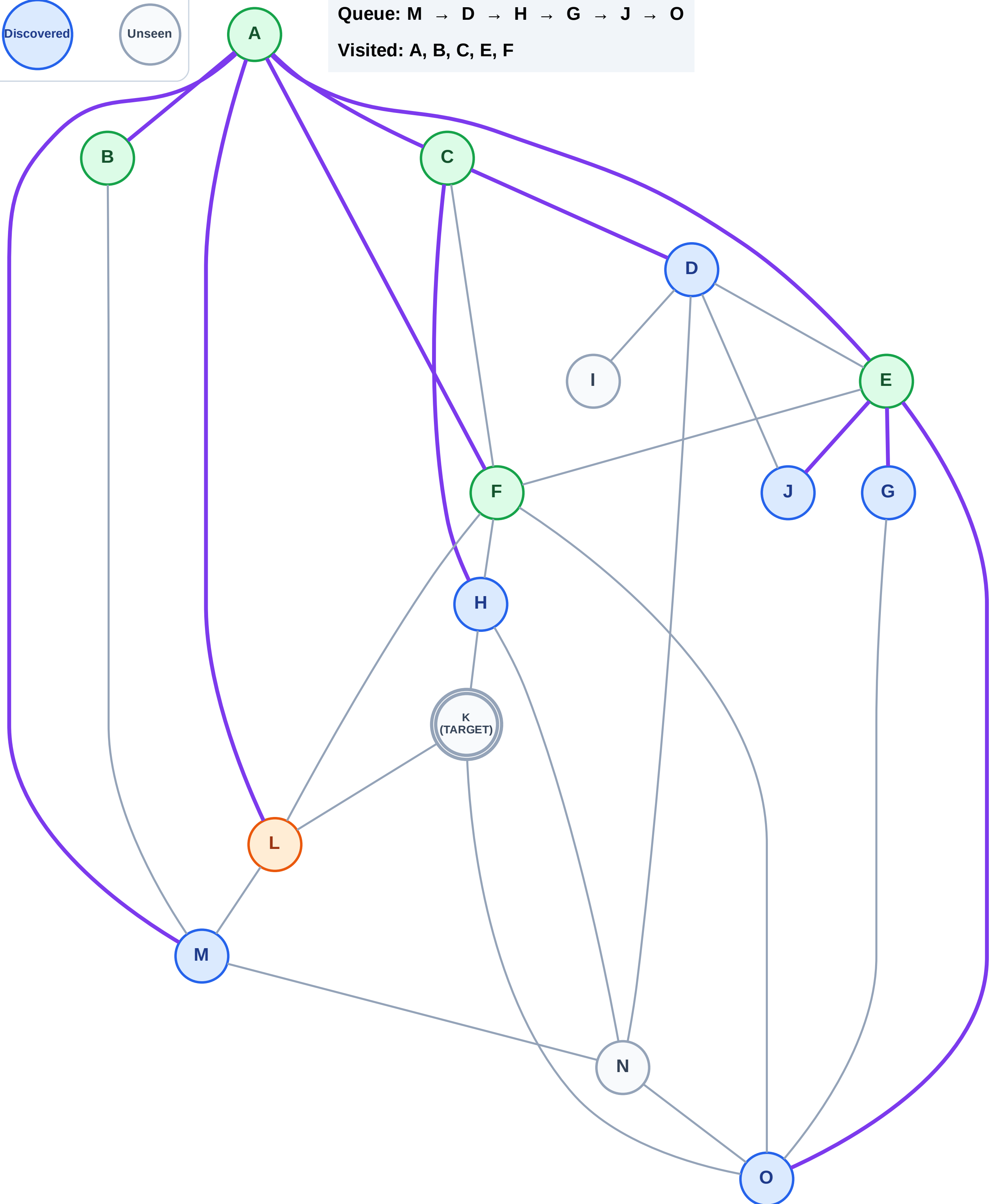
Discovered



Unseen

Queue: M → D → H → G → J → O

Visited: A, B, C, E, F



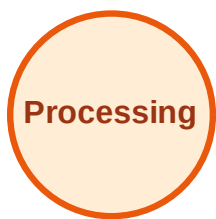
BFS Traversal

Step 13: Discover K from L

Legend



Complete



Processing



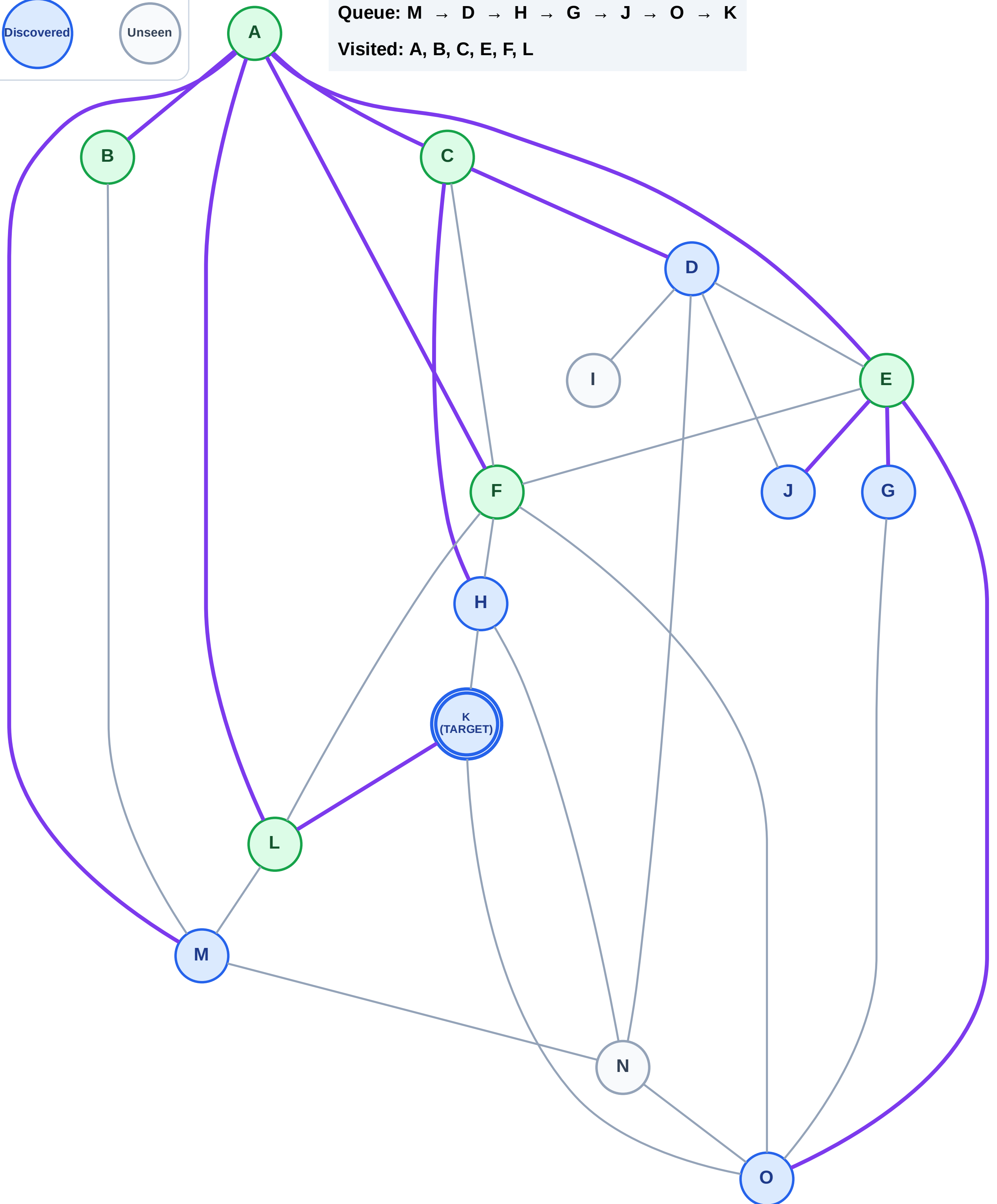
Discovered



Unseen

Queue: M → D → H → G → J → O → K

Visited: A, B, C, E, F, L



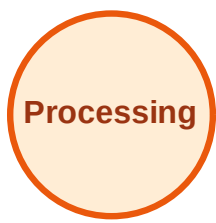
BFS Traversal

Step 14: Process node M

Legend



Complete



Processing



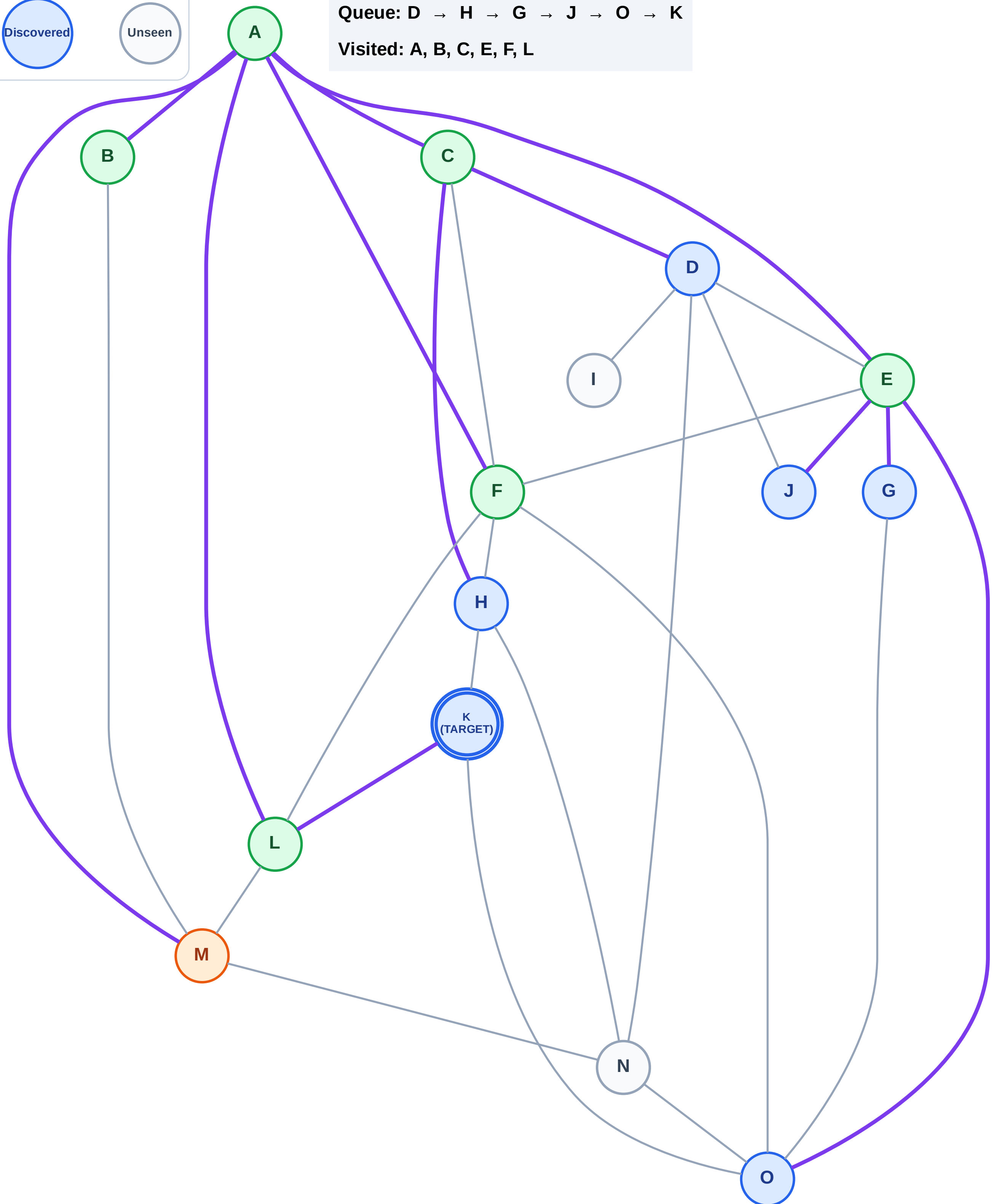
Discovered



Unseen

Queue: D → H → G → J → O → K

Visited: A, B, C, E, F, L



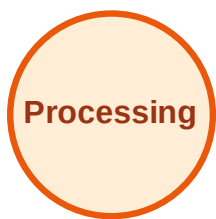
BFS Traversal

Step 15: Discover N from M

Legend



Complete



Processing



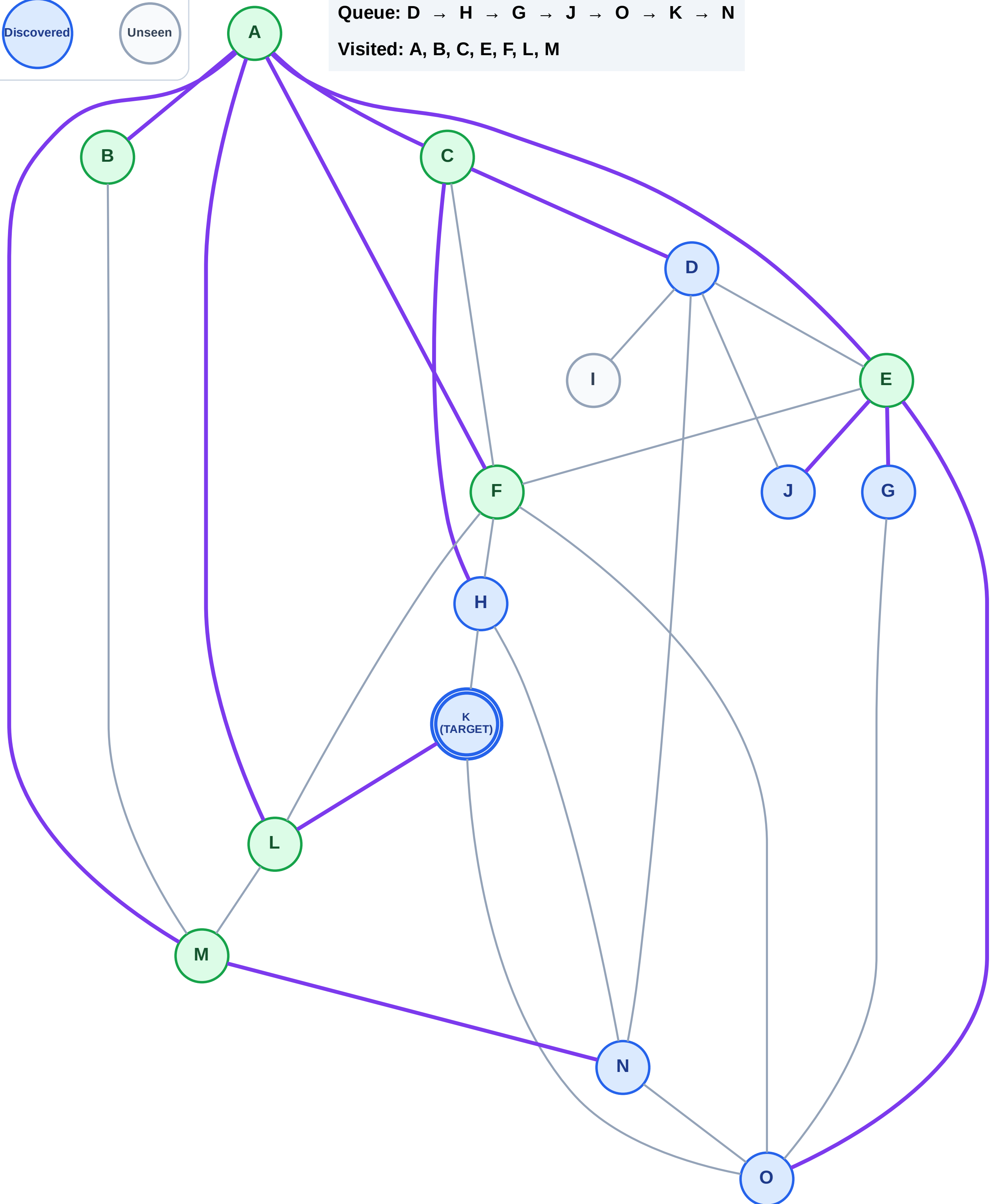
Discovered



Unseen

Queue: D → H → G → J → O → K → N

Visited: A, B, C, E, F, L, M



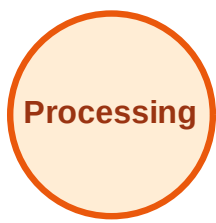
BFS Traversal

Step 16: Process node D

Legend



Complete



Processing



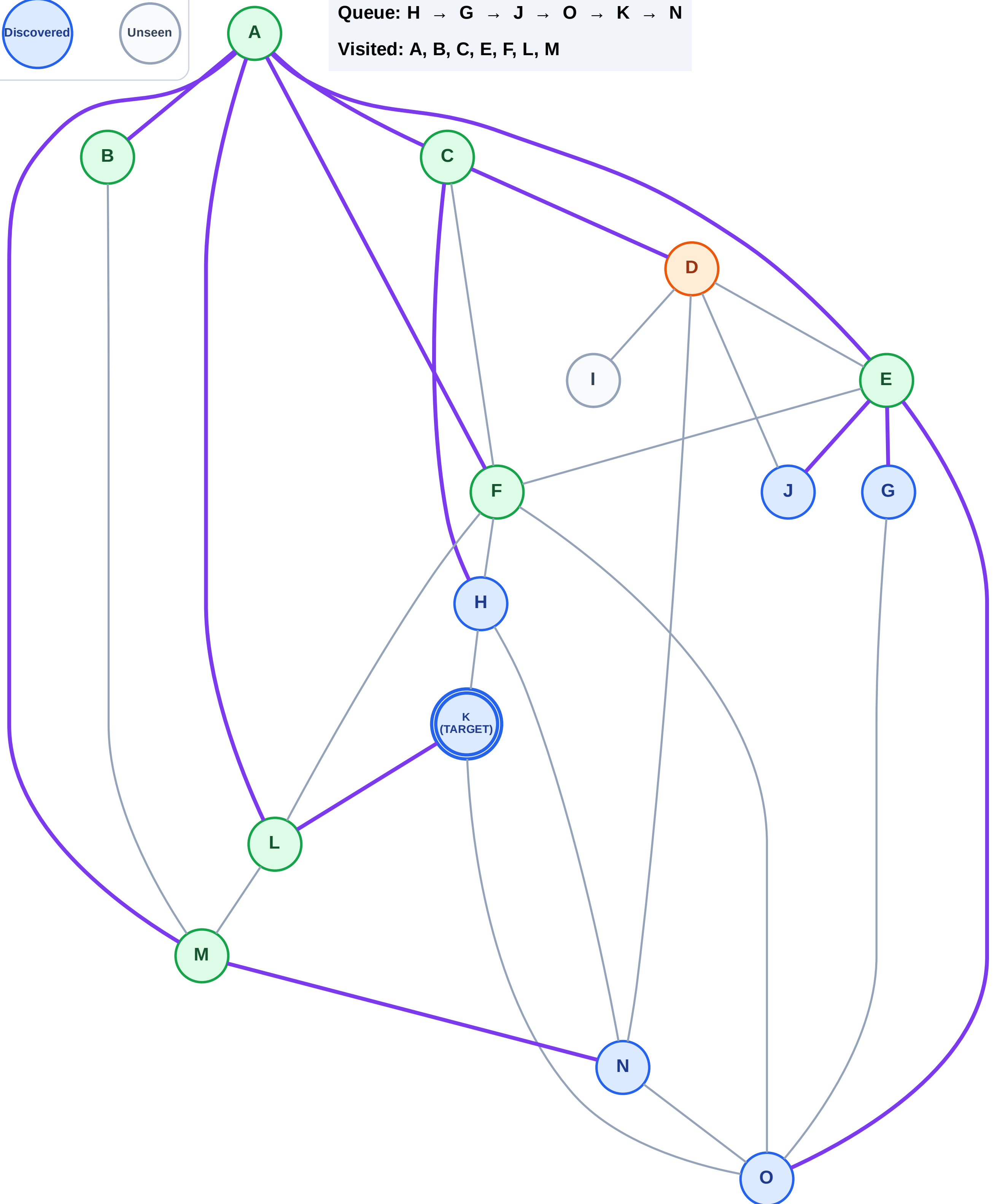
Discovered



Unseen

Queue: H → G → J → O → K → N

Visited: A, B, C, E, F, L, M



BFS Traversal

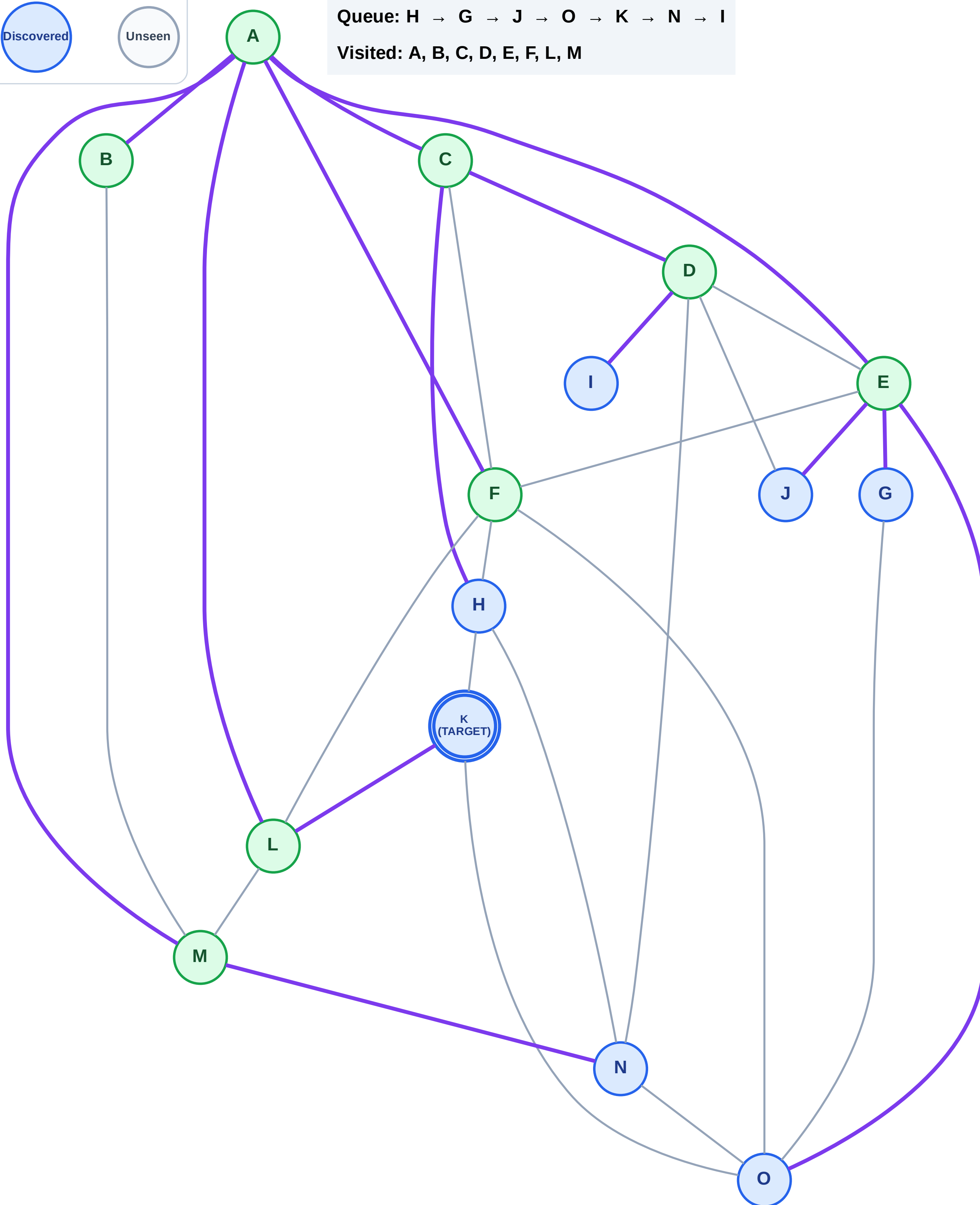
Step 17: Discover I from D

Legend



Queue: H → G → J → O → K → N → I

Visited: A, B, C, D, E, F, L, M



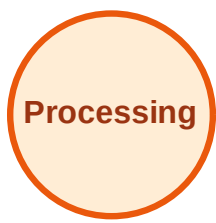
BFS Traversal

Step 18: Process node H

Legend



Complete



Processing



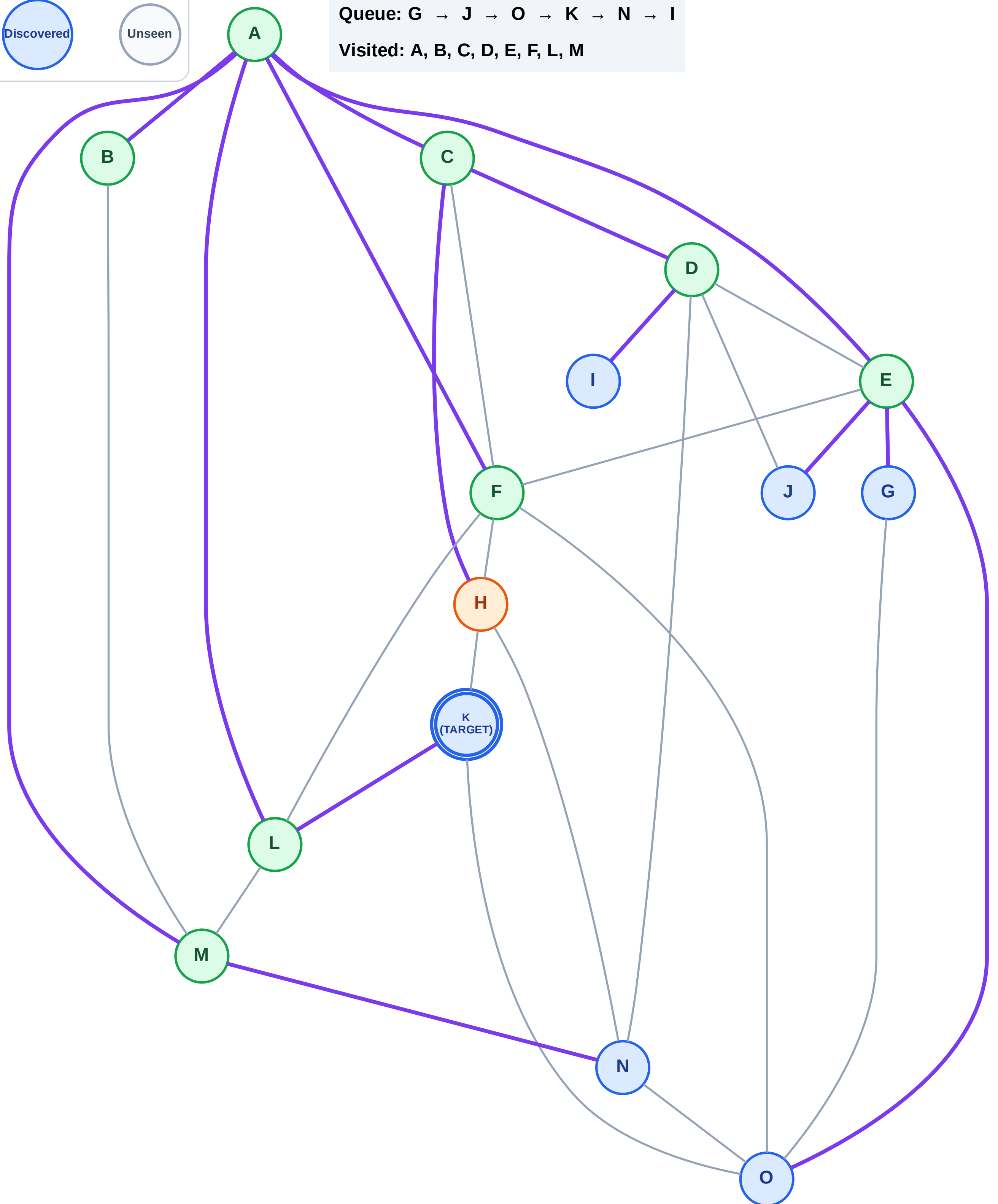
Discovered



Unseen

Queue: G → J → O → K → N → I

Visited: A, B, C, D, E, F, L, M



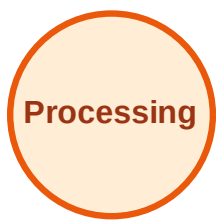
BFS Traversal

Step 19: No new neighbors from H

Legend



Complete



Processing



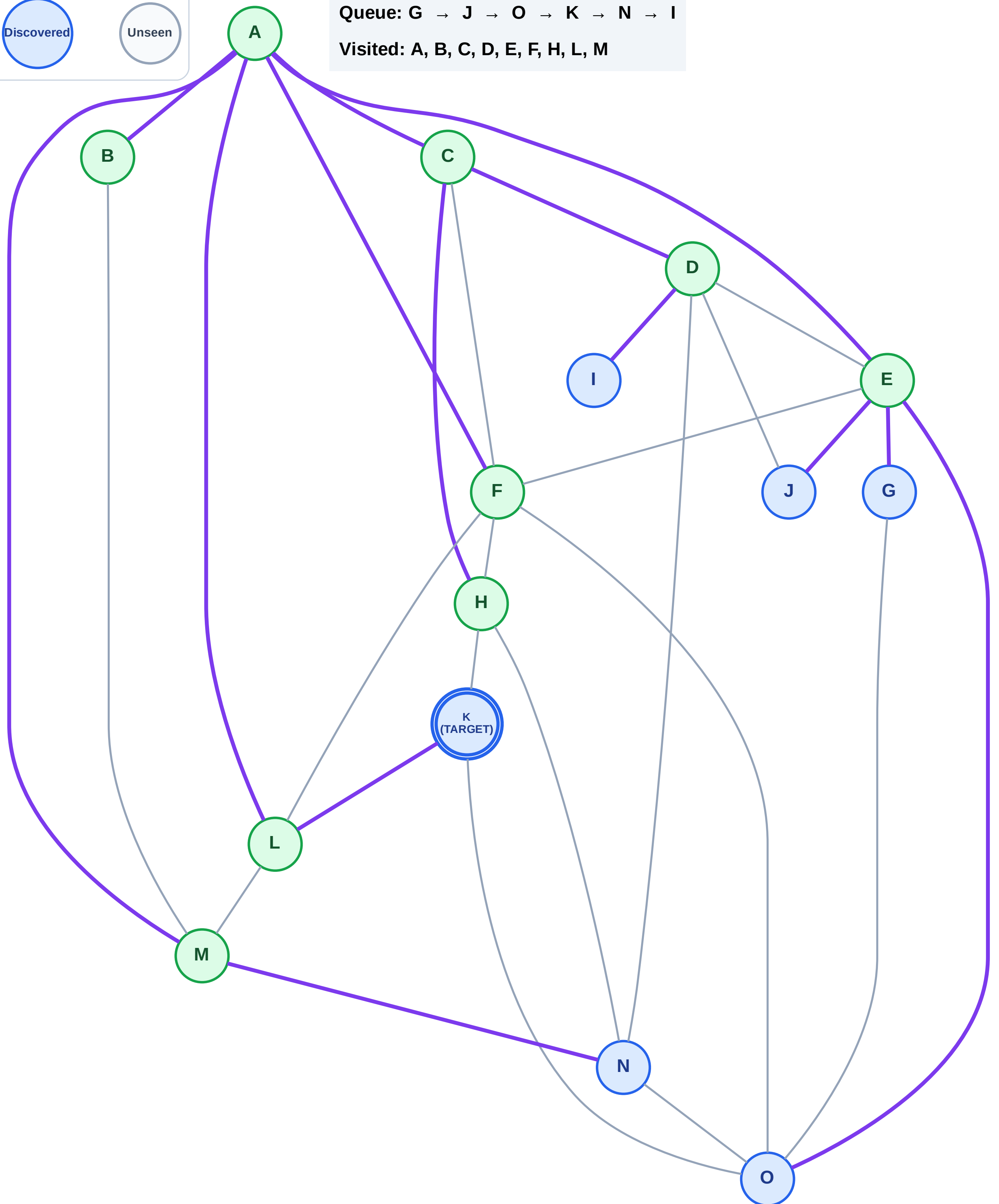
Discovered



Unseen

Queue: G → J → O → K → N → I

Visited: A, B, C, D, E, F, H, L, M



BFS Traversal

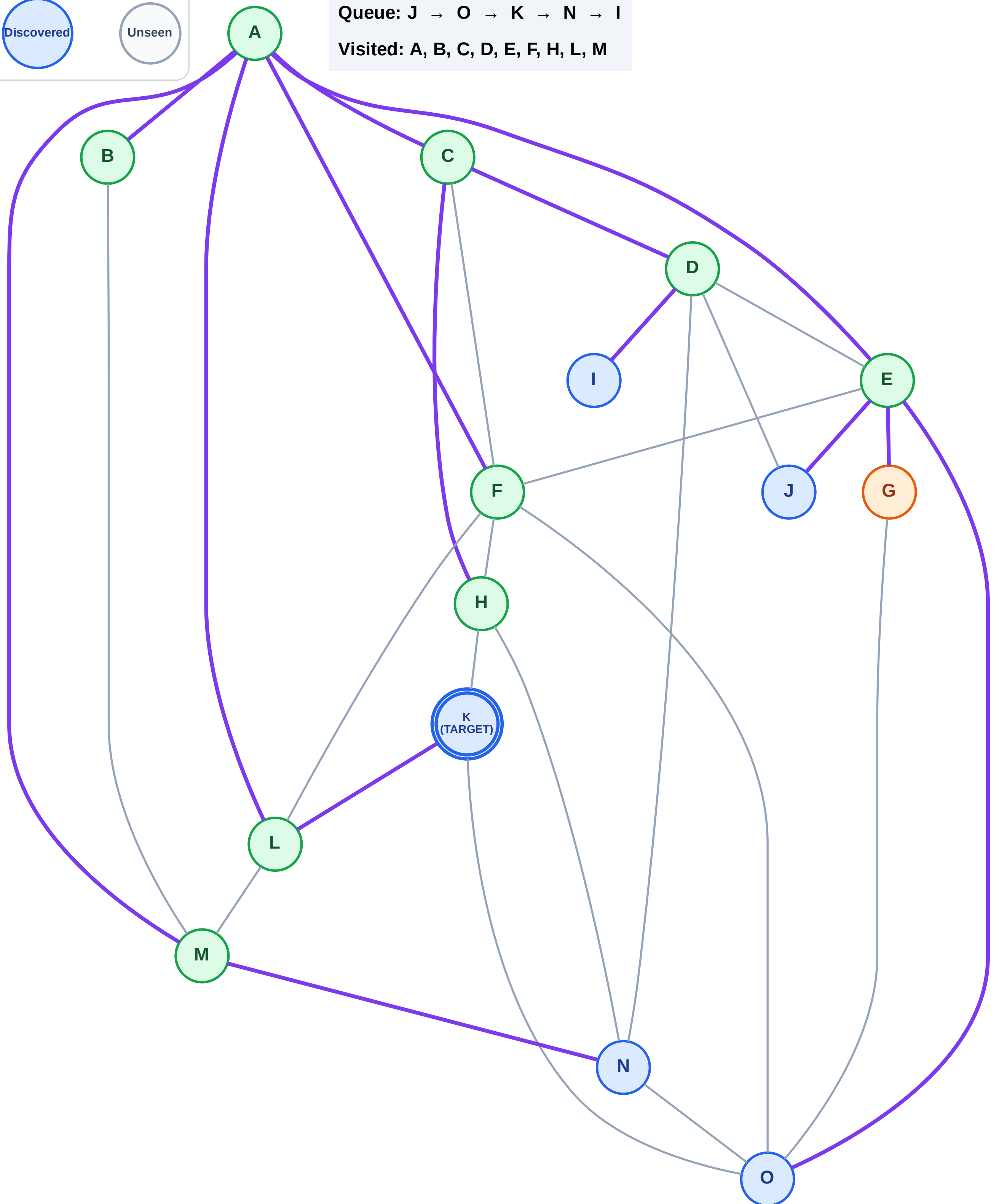
Step 20: Process node G

Legend



Queue: J → O → K → N → I

Visited: A, B, C, D, E, F, H, L, M



BFS Traversal

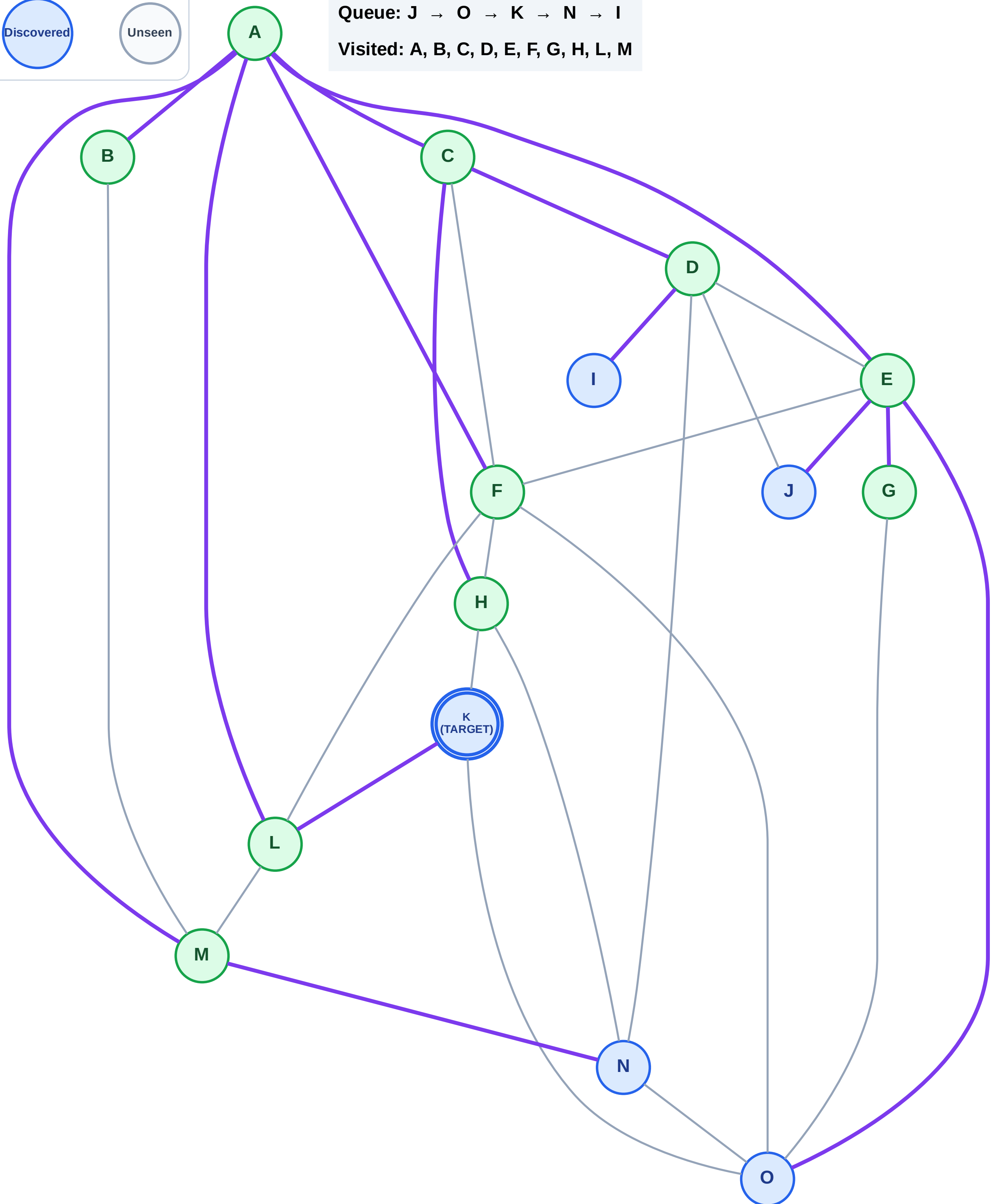
Step 21: No new neighbors from G

Legend



Queue: J → O → K → N → I

Visited: A, B, C, D, E, F, G, H, L, M



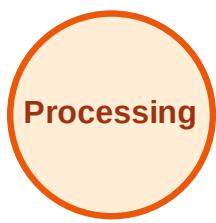
BFS Traversal

Step 22: Process node J

Legend



Complete



Processing



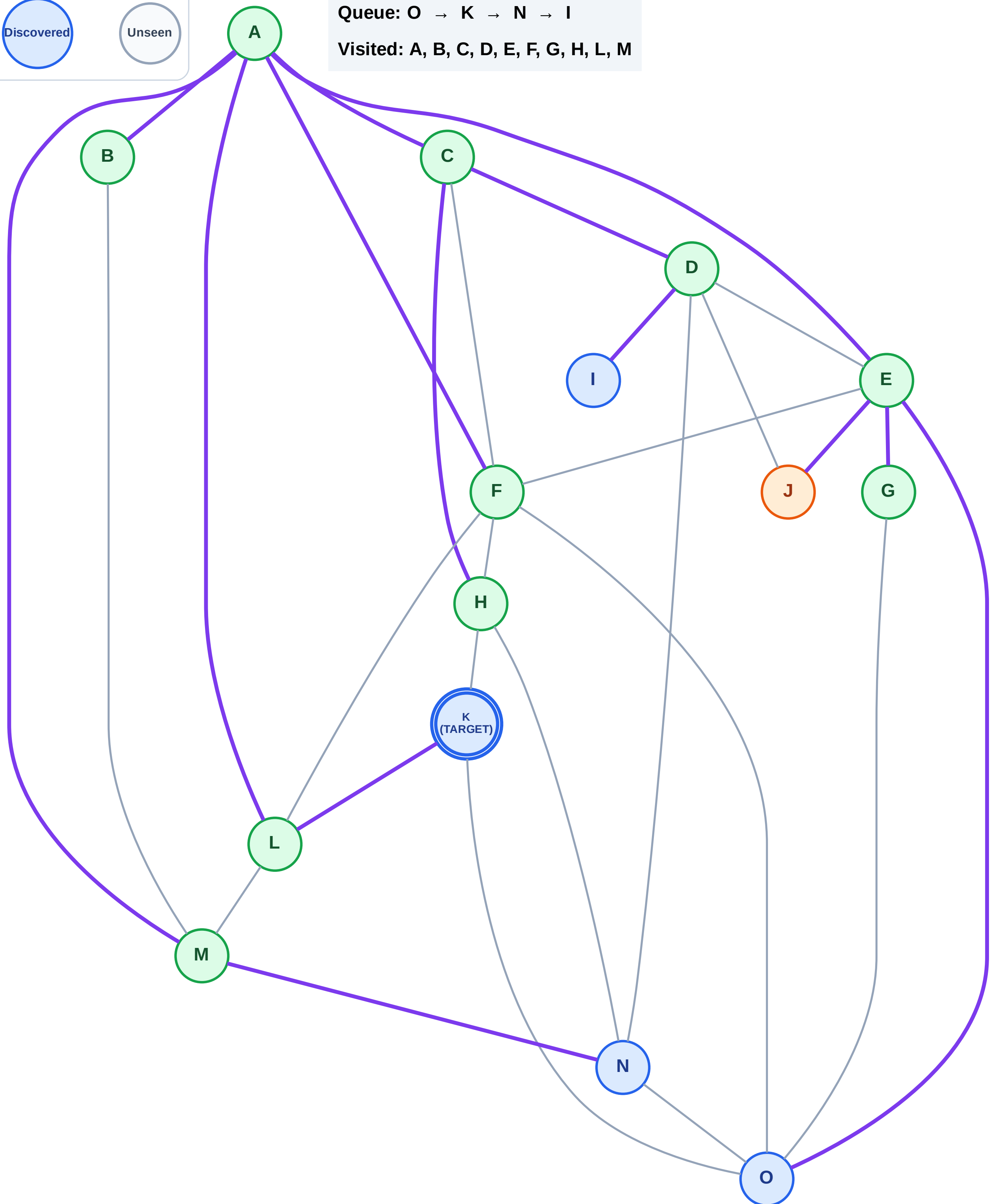
Discovered



Unseen

Queue: O → K → N → I

Visited: A, B, C, D, E, F, G, H, L, M



BFS Traversal

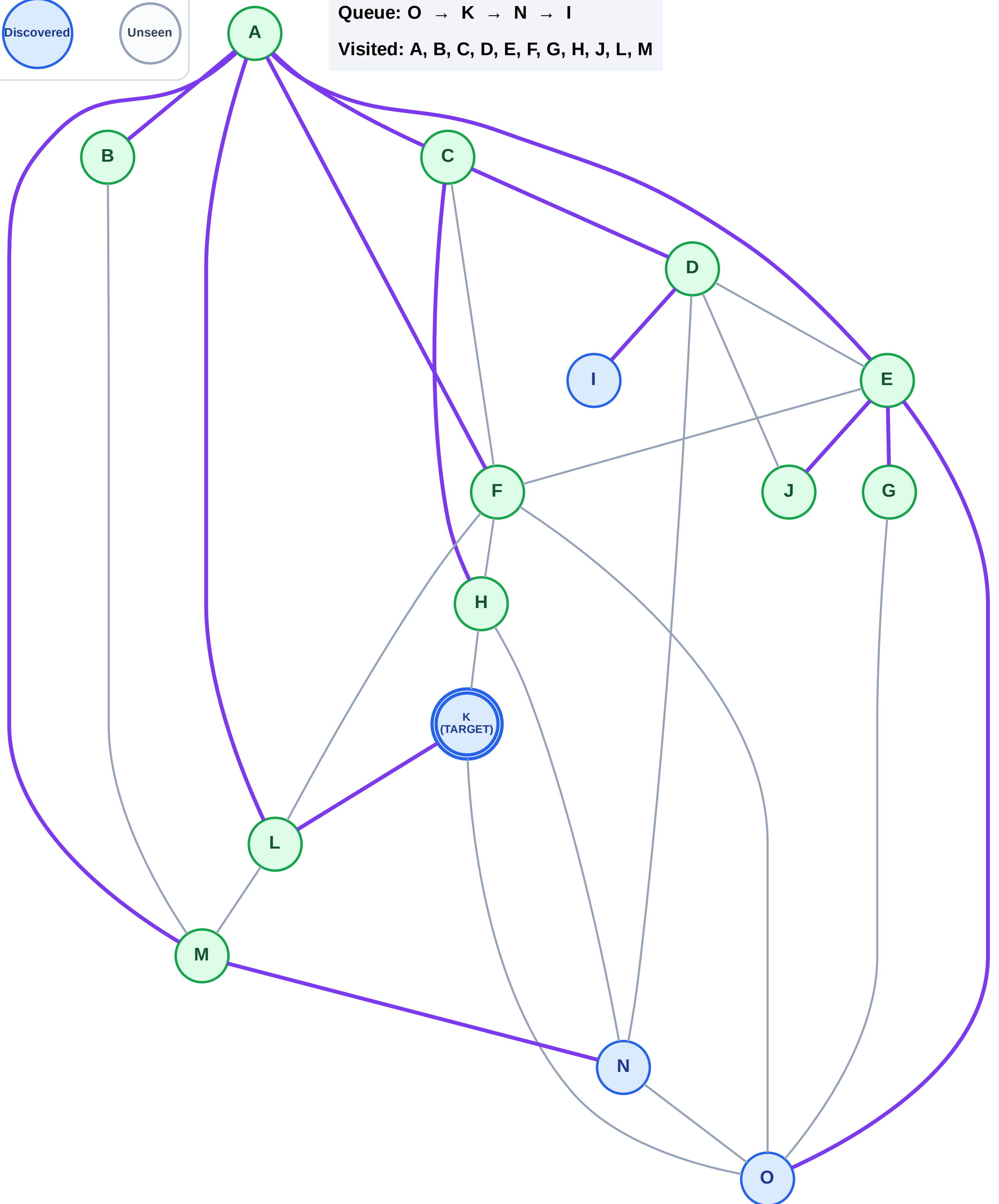
Step 23: No new neighbors from J

Legend



Queue: O → K → N → I

Visited: A, B, C, D, E, F, G, H, J, L, M



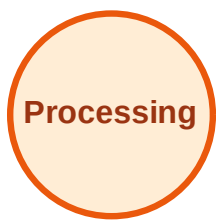
BFS Traversal

Step 24: Process node O

Legend



Complete



Processing



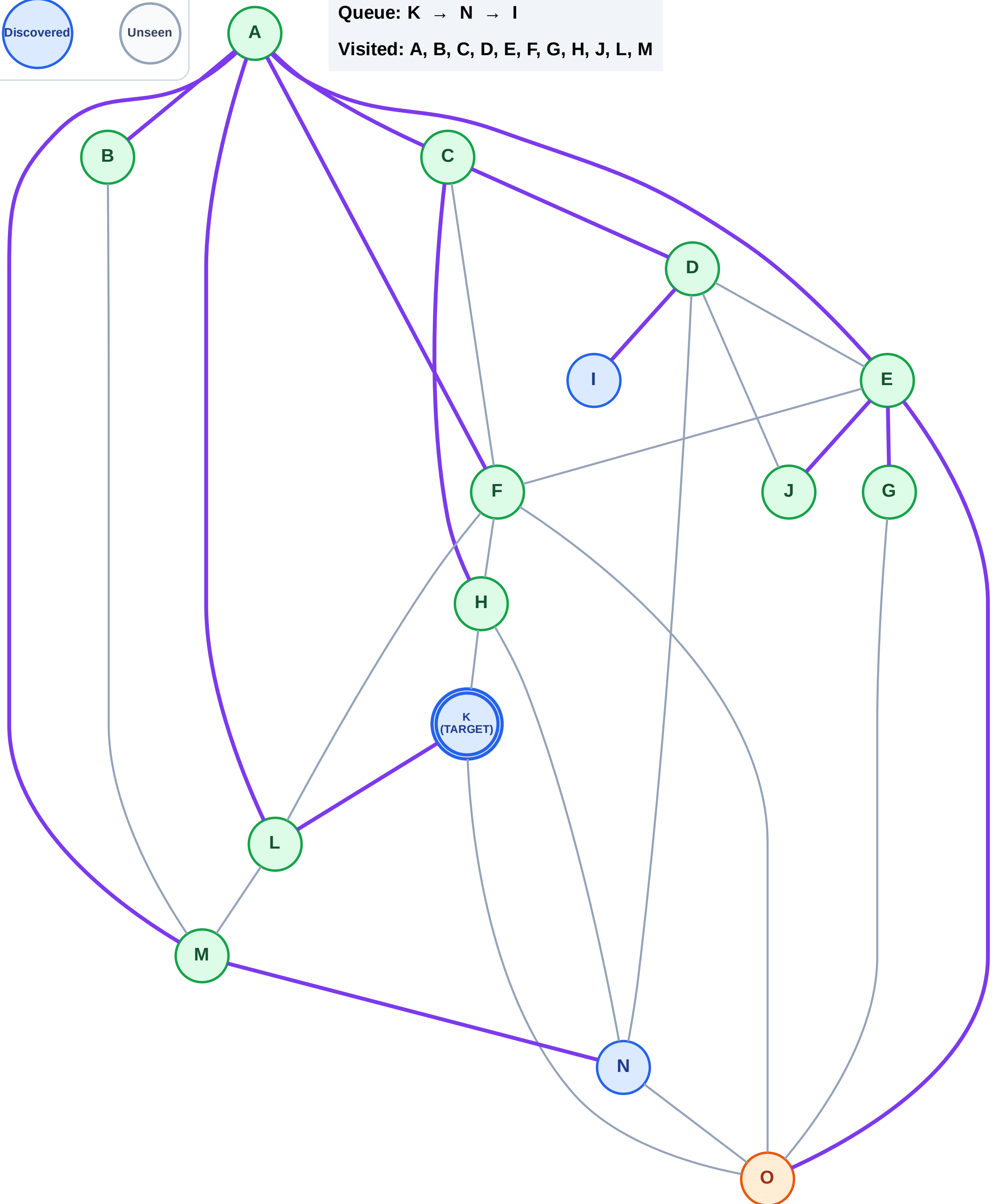
Discovered



Unseen

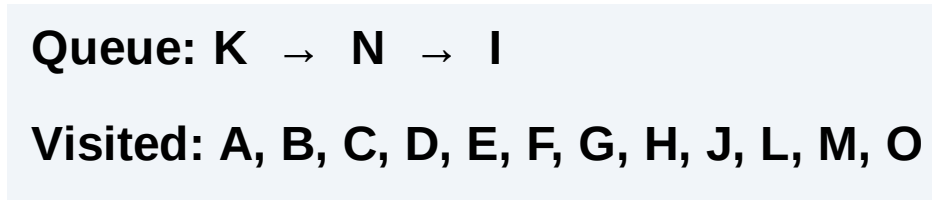
Queue: K → N → I

Visited: A, B, C, D, E, F, G, H, J, L, M



Step 25: No new neighbors from O

Step 25: No new neighbors from O



BFS Traversal

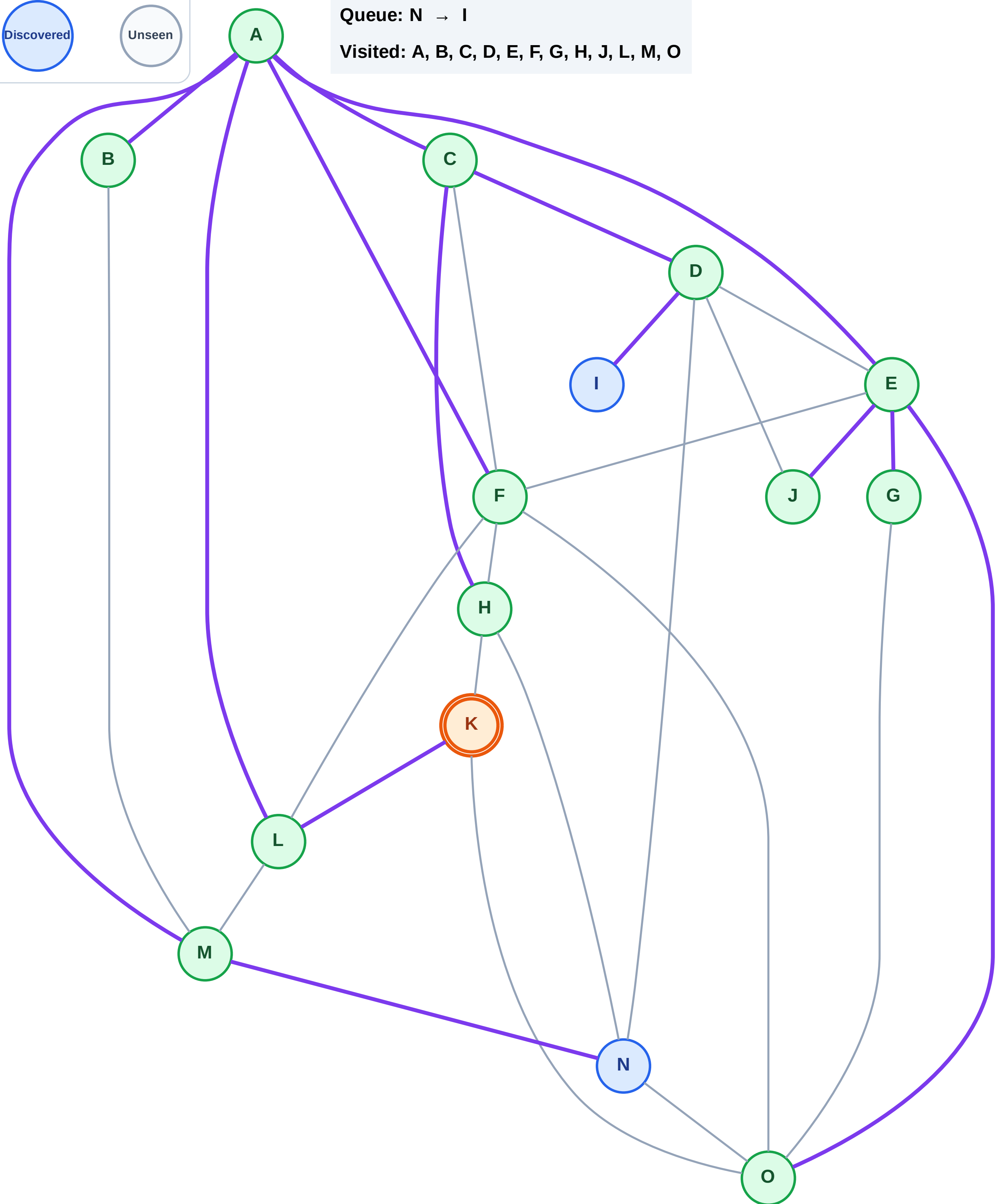
Step 26: Process node K

Legend



Queue: N → I

Visited: A, B, C, D, E, F, G, H, J, L, M, O



BFS Traversal

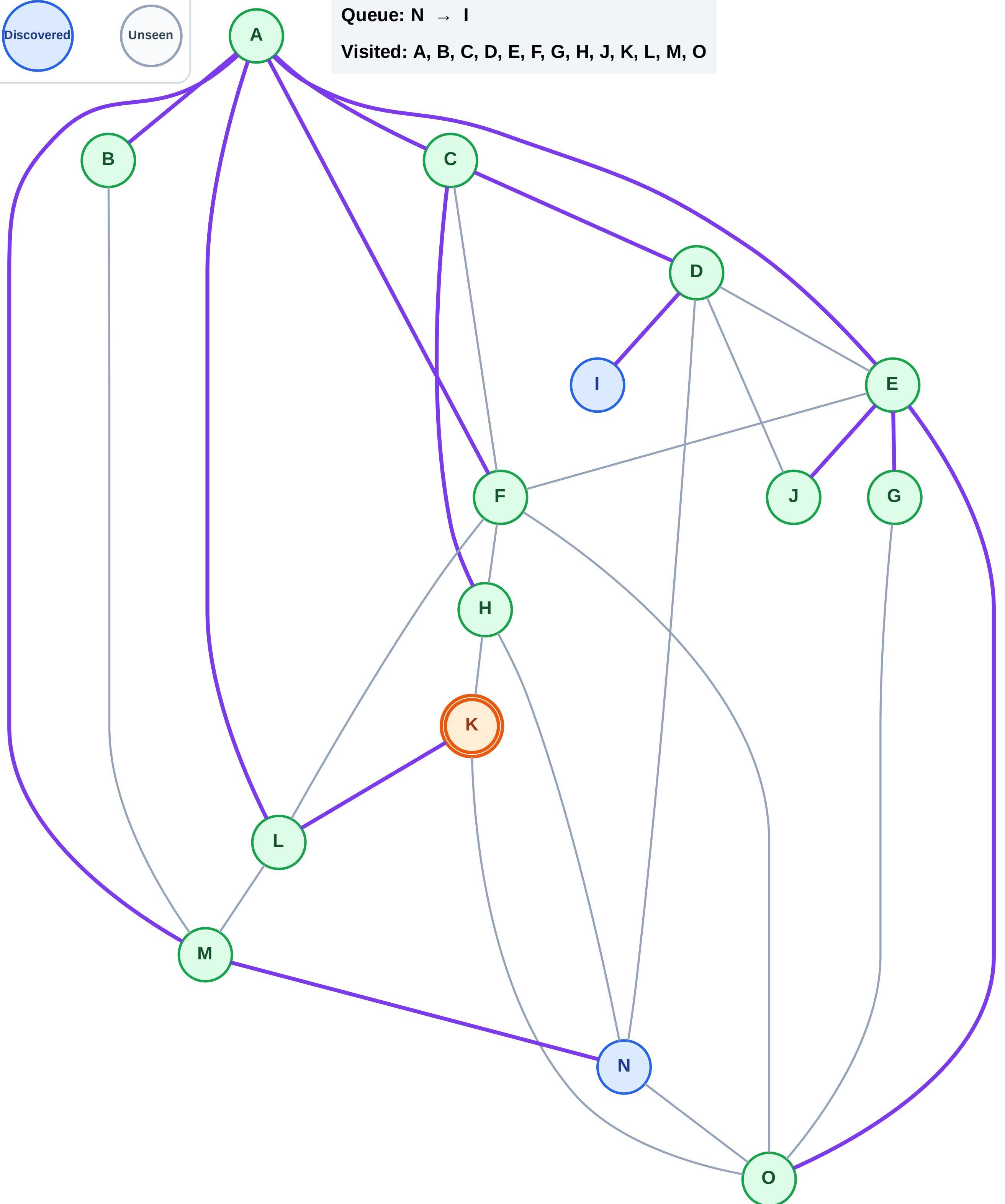
Step 27: TARGET NODE K REACHED! Path (2 hops): A → L → K

Legend



Queue: N → I

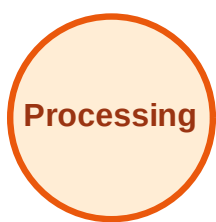
Visited: A, B, C, D, E, F, G, H, J, K, L, M, O



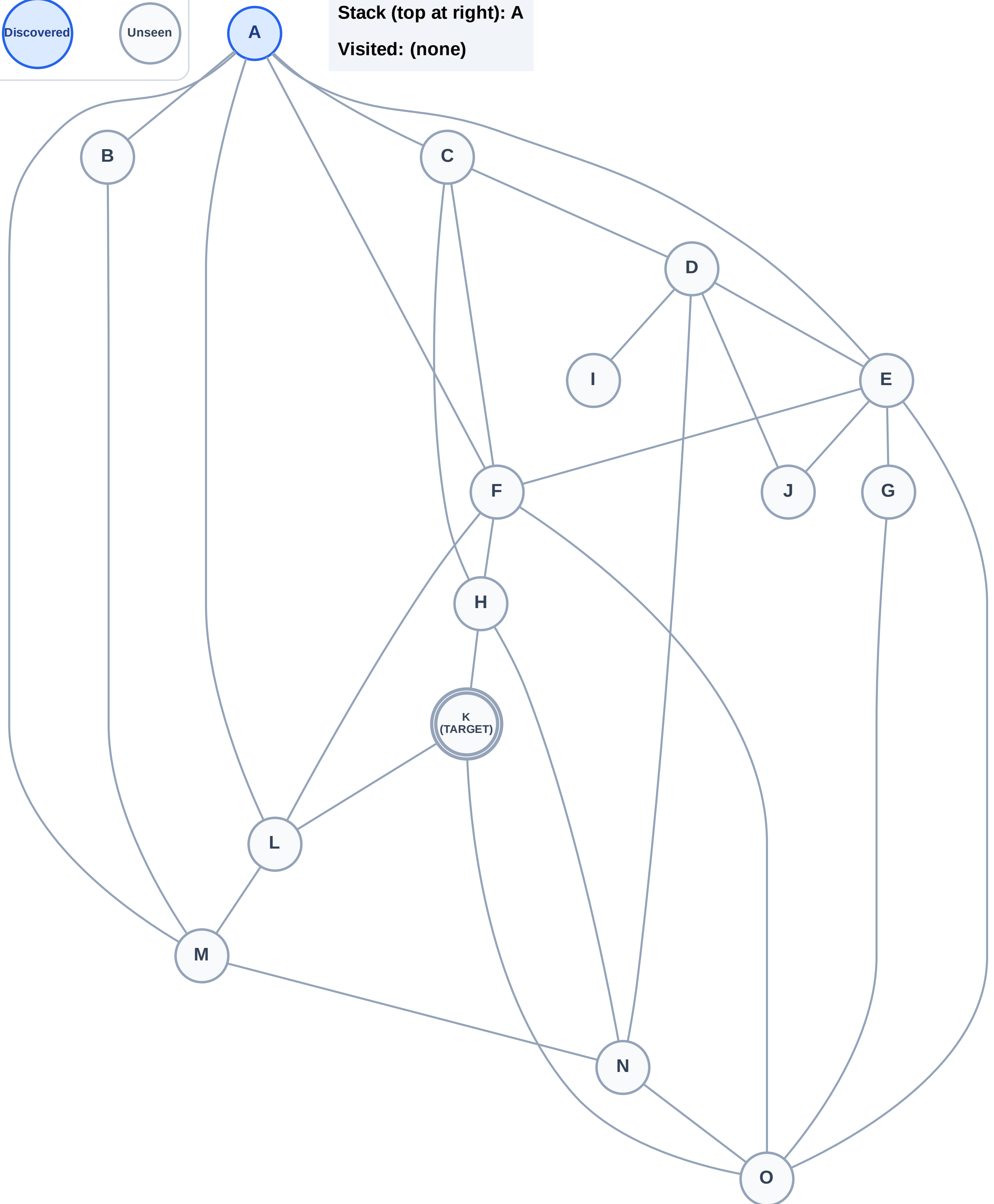
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

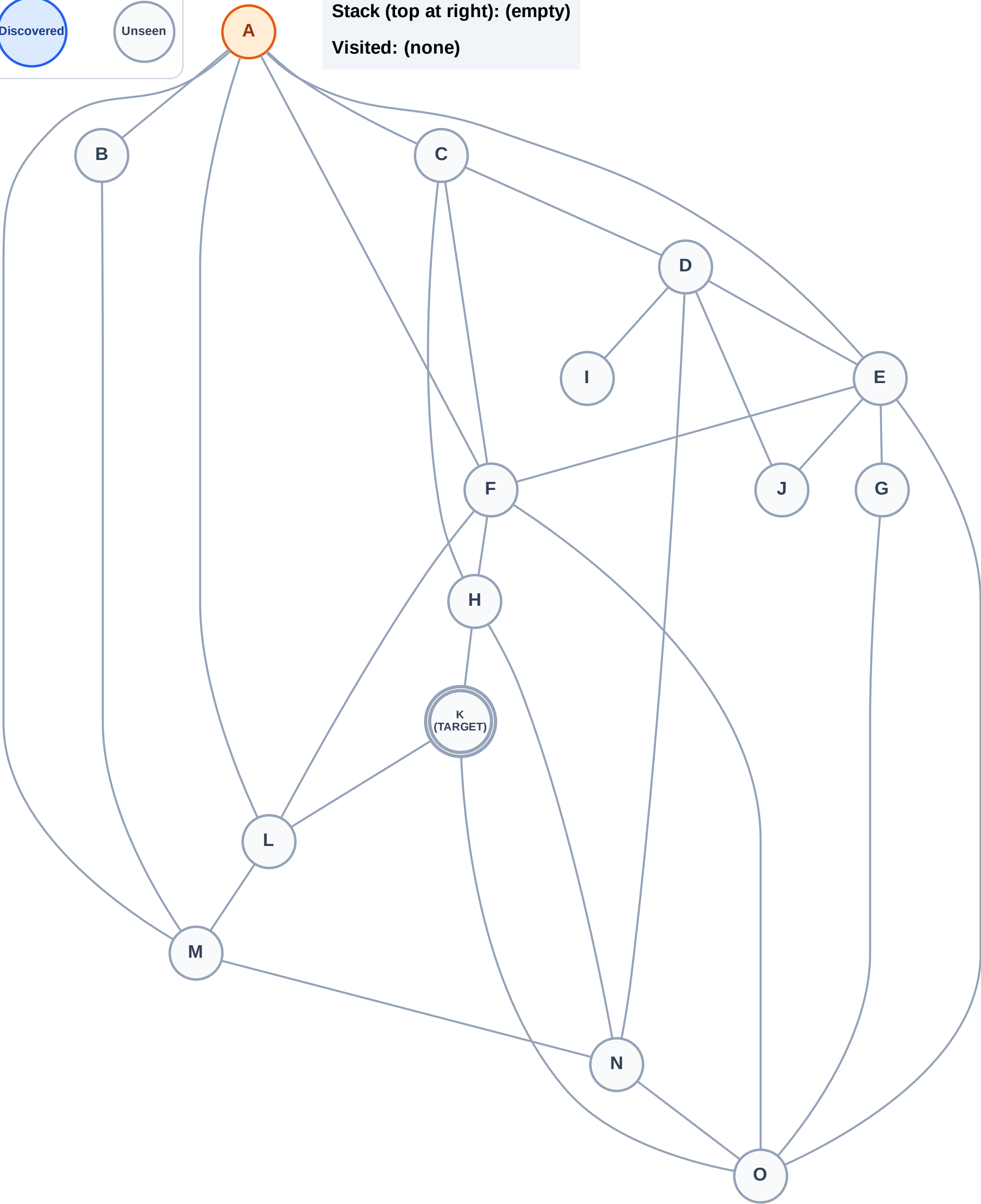
Complete

Processing

Discovered

Unseen

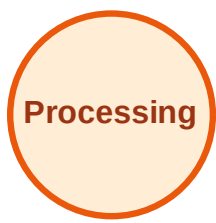
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

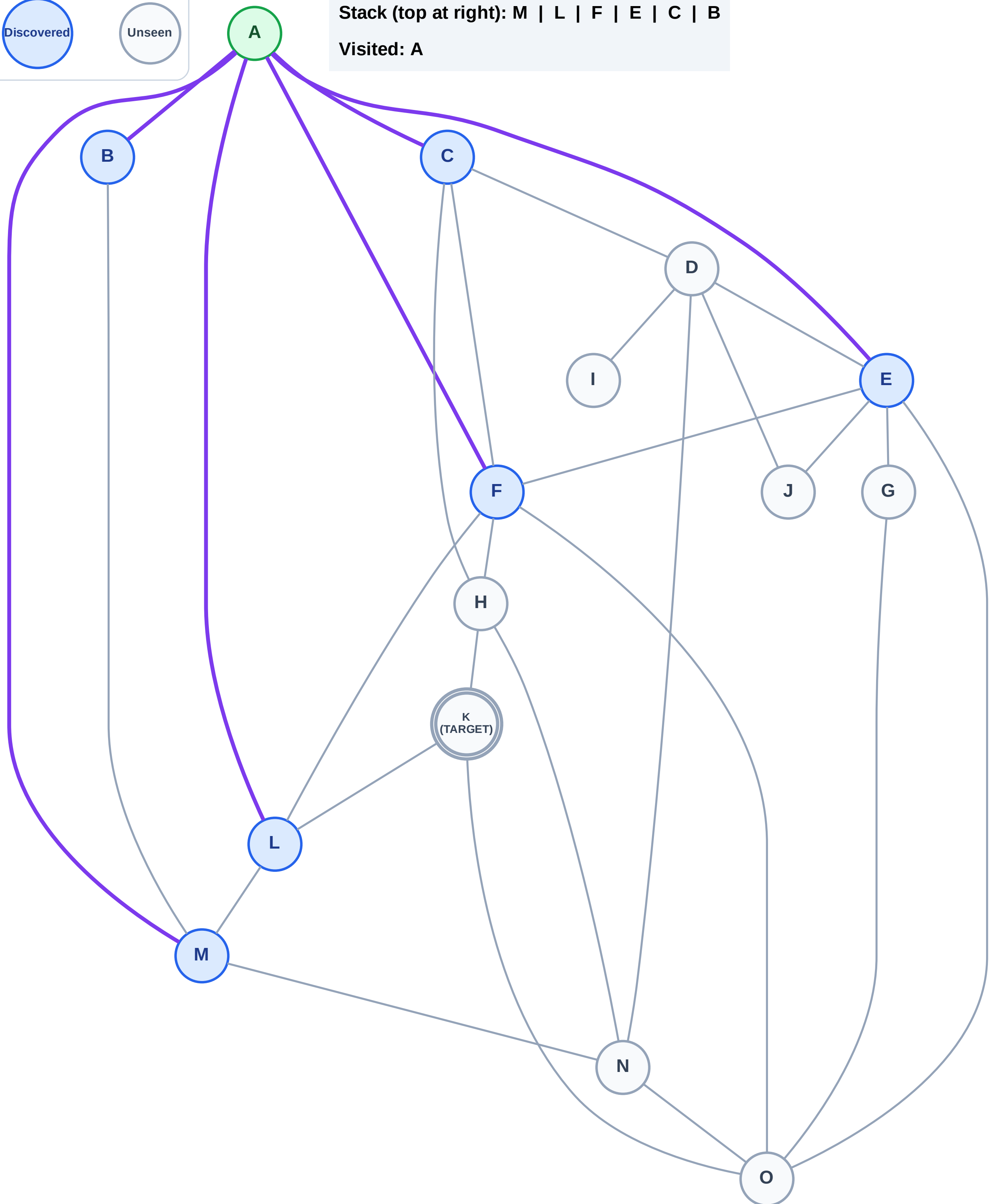
Step 3: Discover M, L, F, E, C, B from A

Legend



Stack (top at right): M | L | F | E | C | B

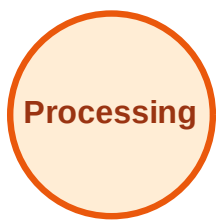
Visited: A



DFS Traversal

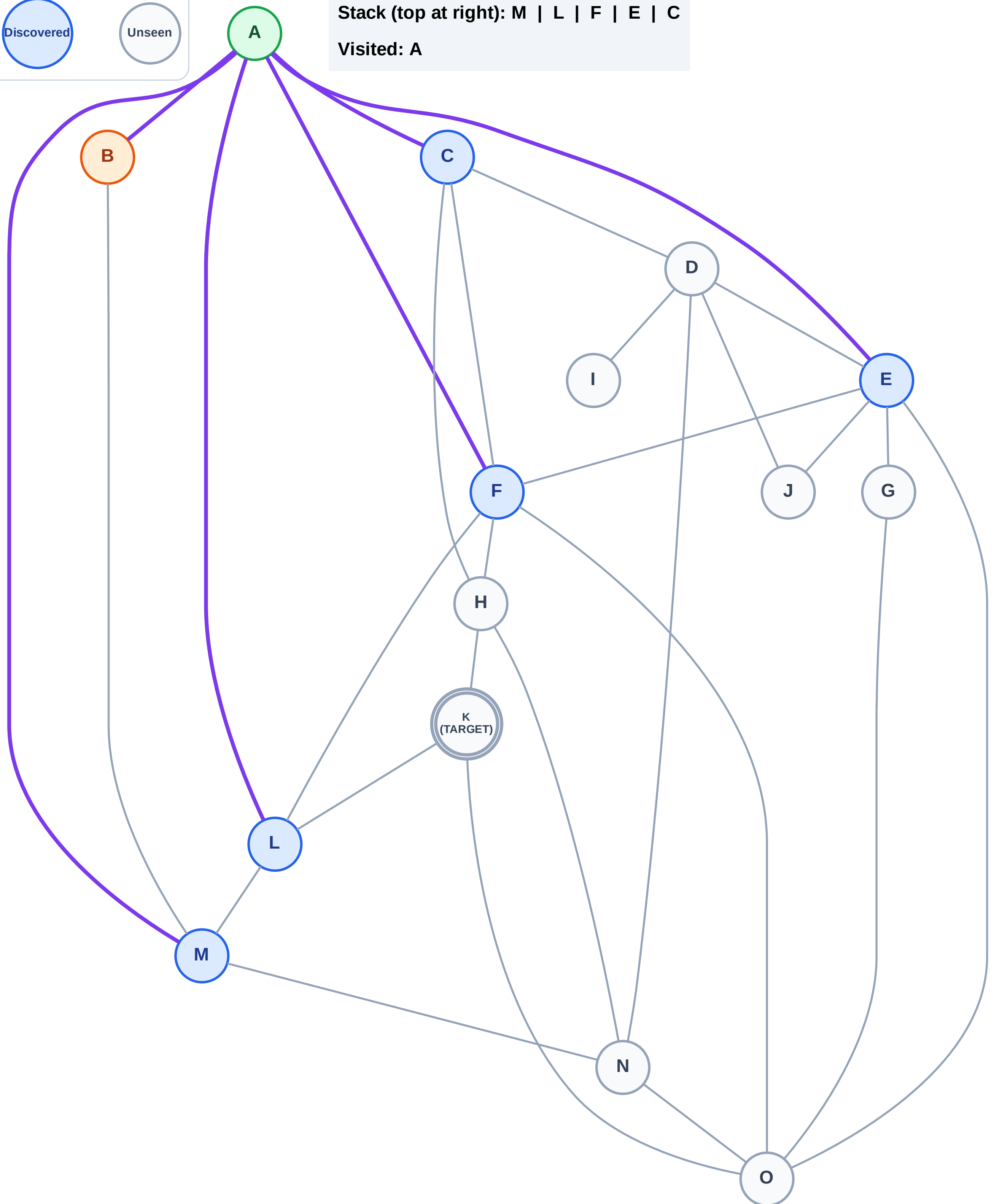
Step 4: Process node B

Legend



Stack (top at right): M | L | F | E | C

Visited: A

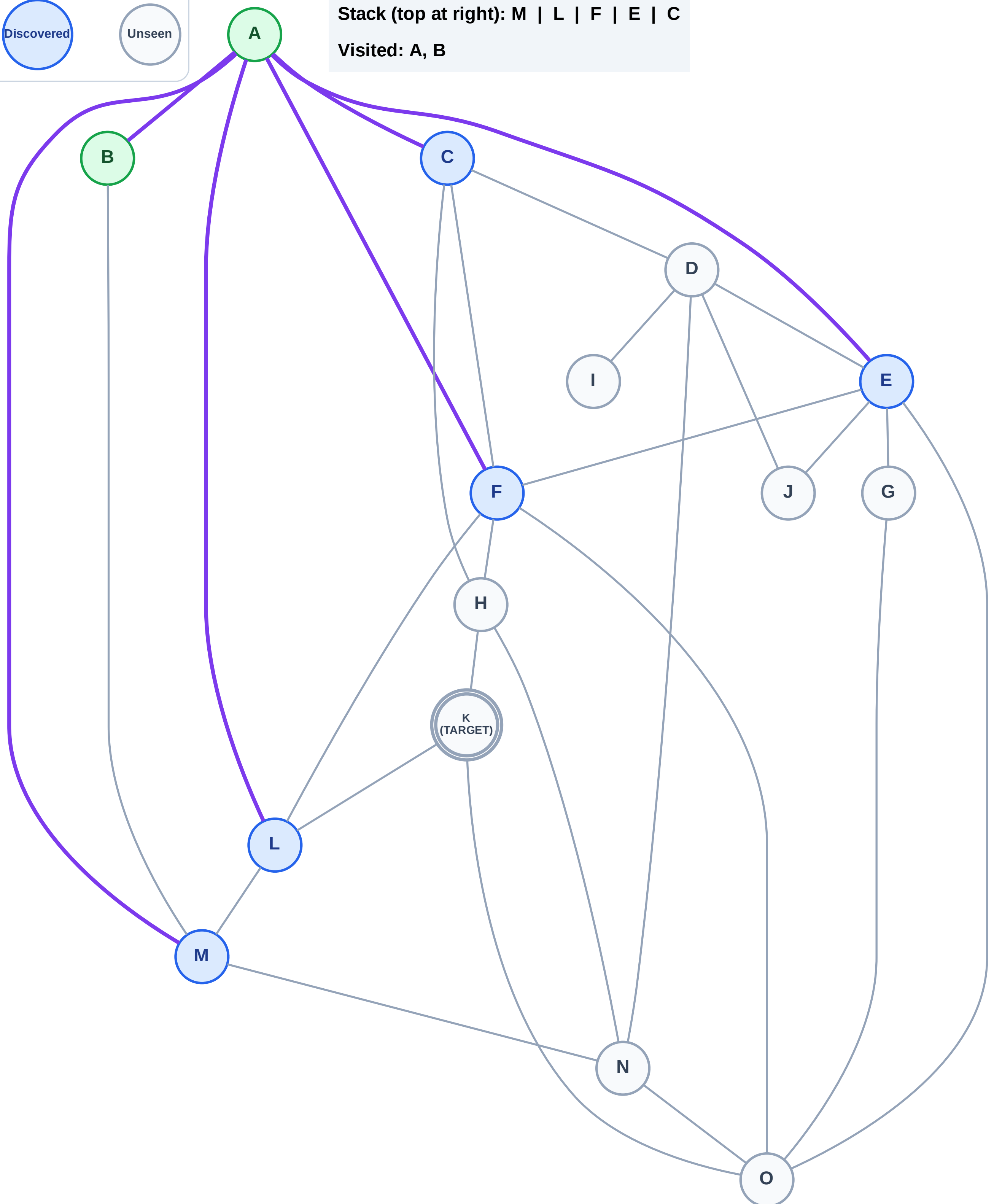


Step 5: No new neighbors from B

Stack (top at right): M | L | F | E | C

Legend

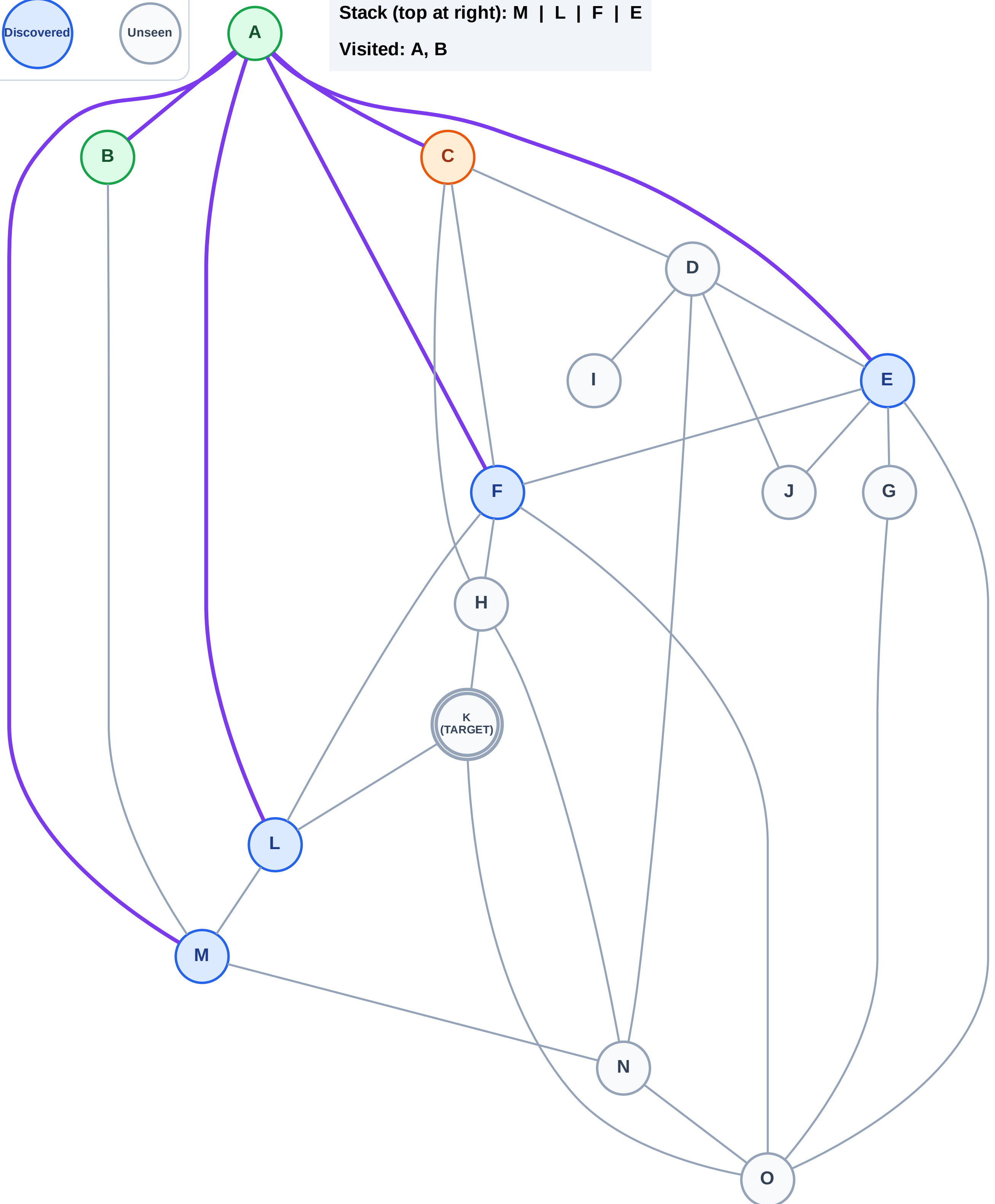
Unseen



Step 6: Process node C



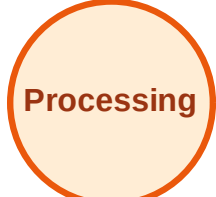
Visited: A, B



DFS Traversal

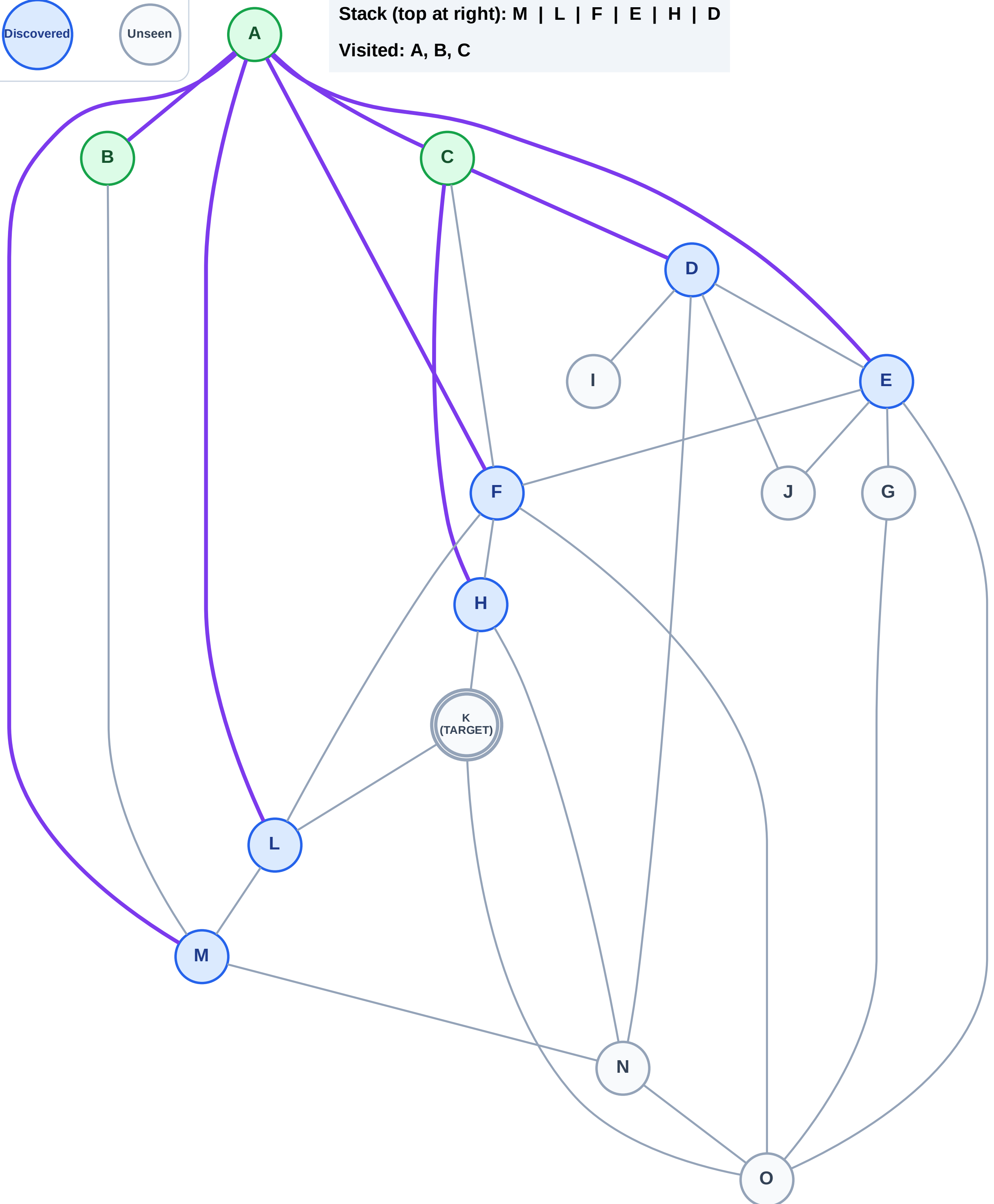
Step 7: Discover H, D from C

Legend



Stack (top at right): M | L | F | E | H | D

Visited: A, B, C



DFS Traversal

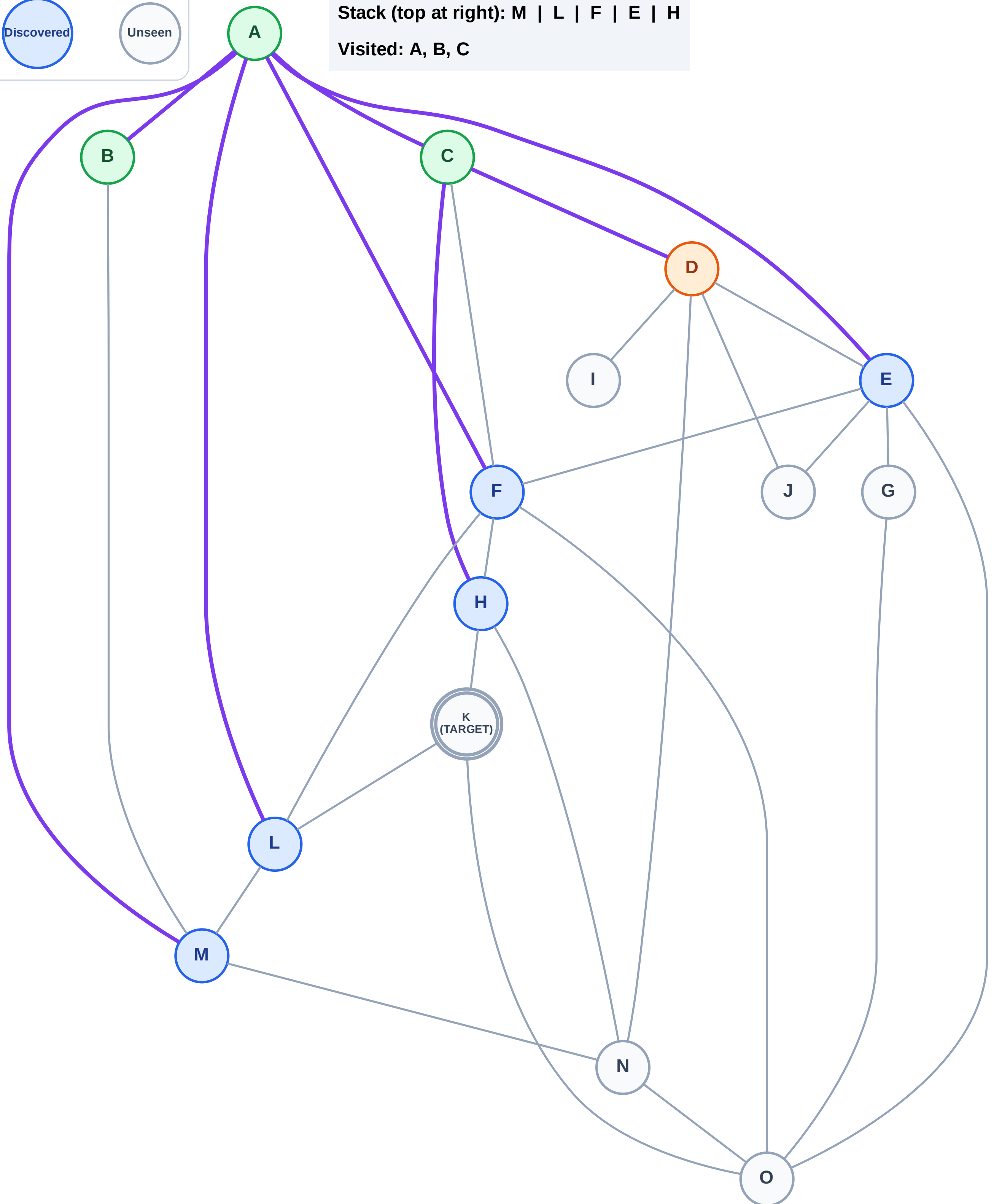
Step 8: Process node D

Legend



Stack (top at right): M | L | F | E | H

Visited: A, B, C



DFS Traversal

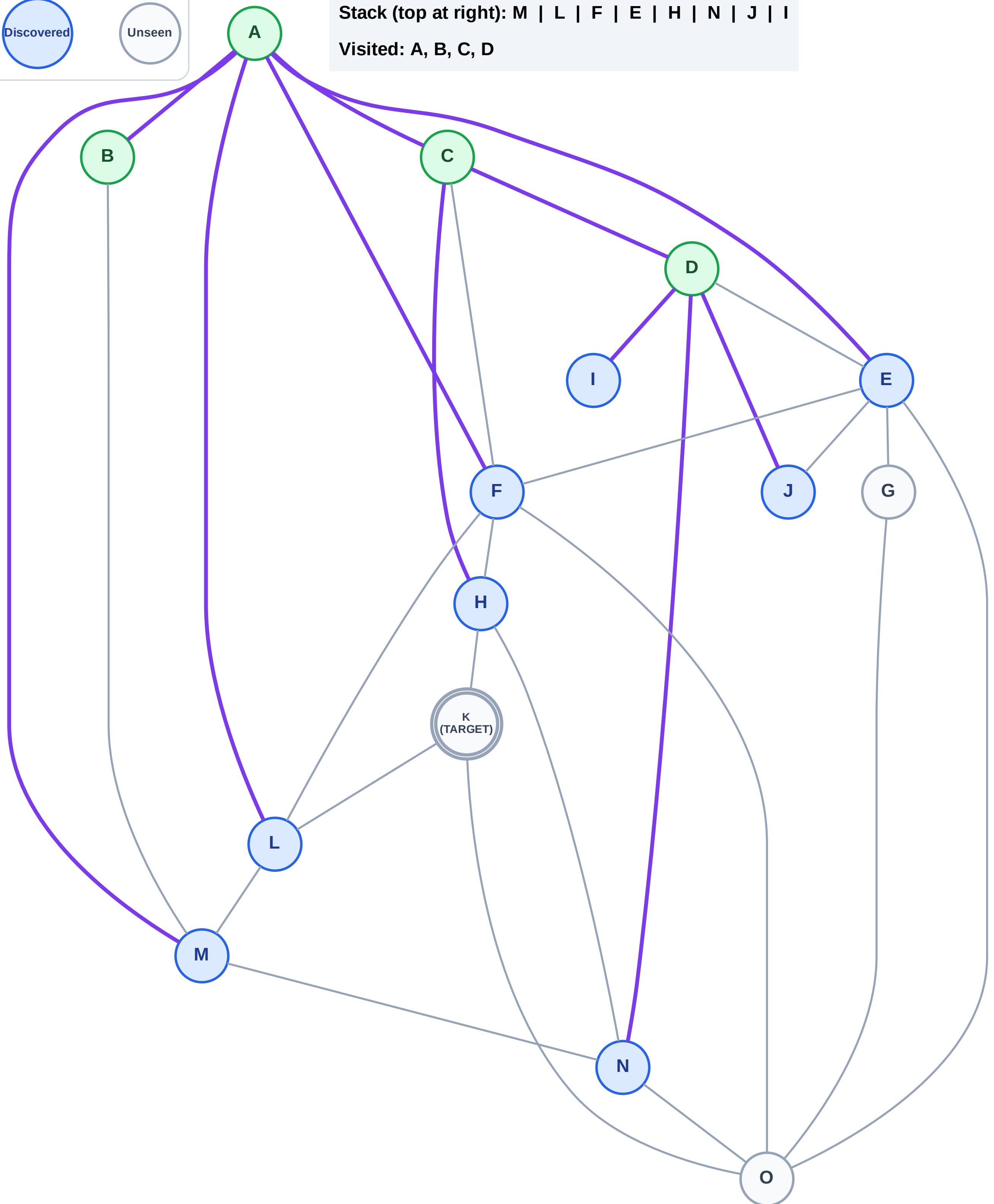
Step 9: Discover N, J, I from D

Legend



Stack (top at right): M | L | F | E | H | N | J | I

Visited: A, B, C, D



DFS Traversal

Step 10: Process node I

Legend

Complete

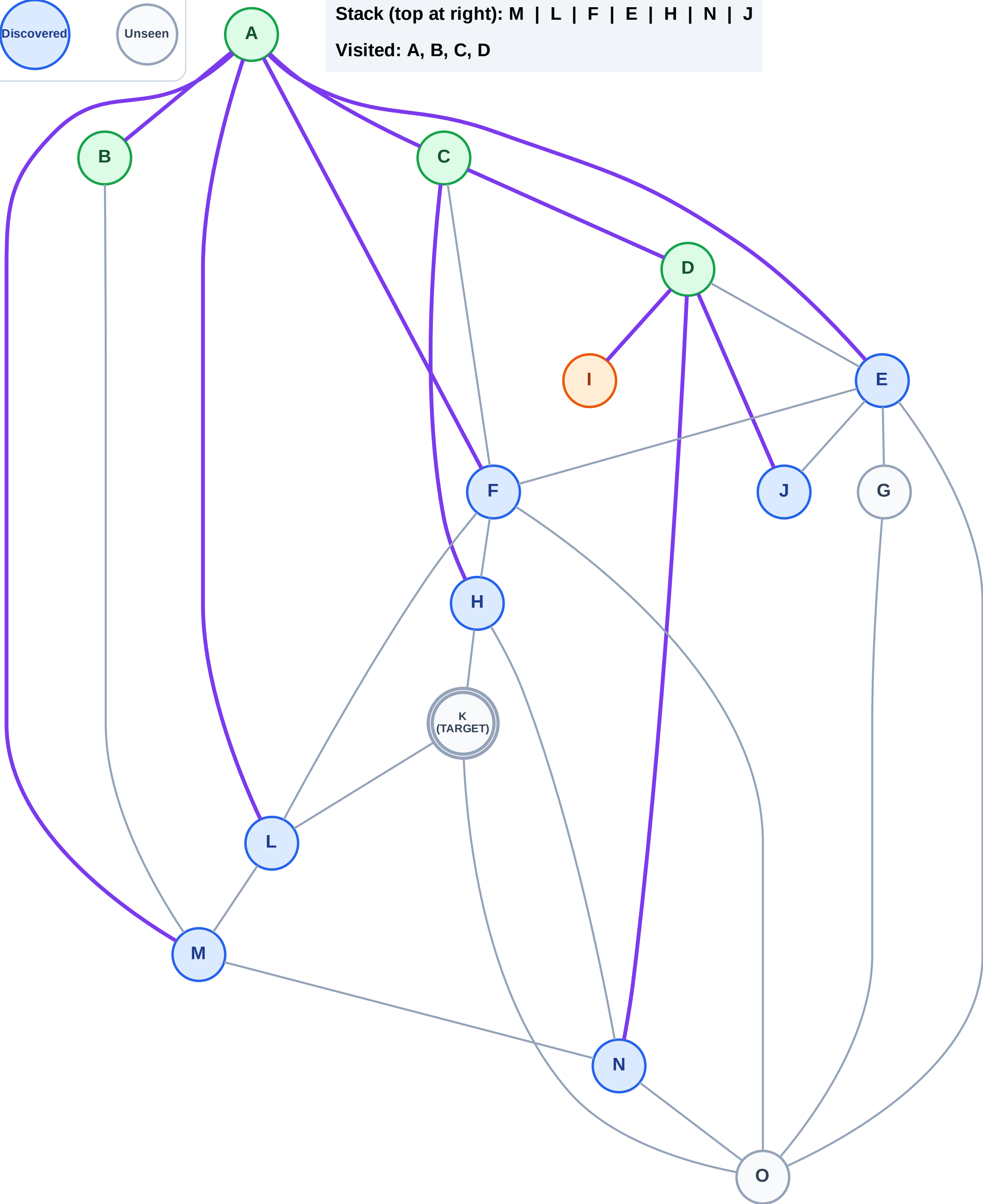
Processing

Discovered

Unseen

Stack (top at right): M | L | F | E | H | N | J

Visited: A, B, C, D



Step 11: No new neighbors from I

Stack (top at right): M | L | F | E | H | N | J

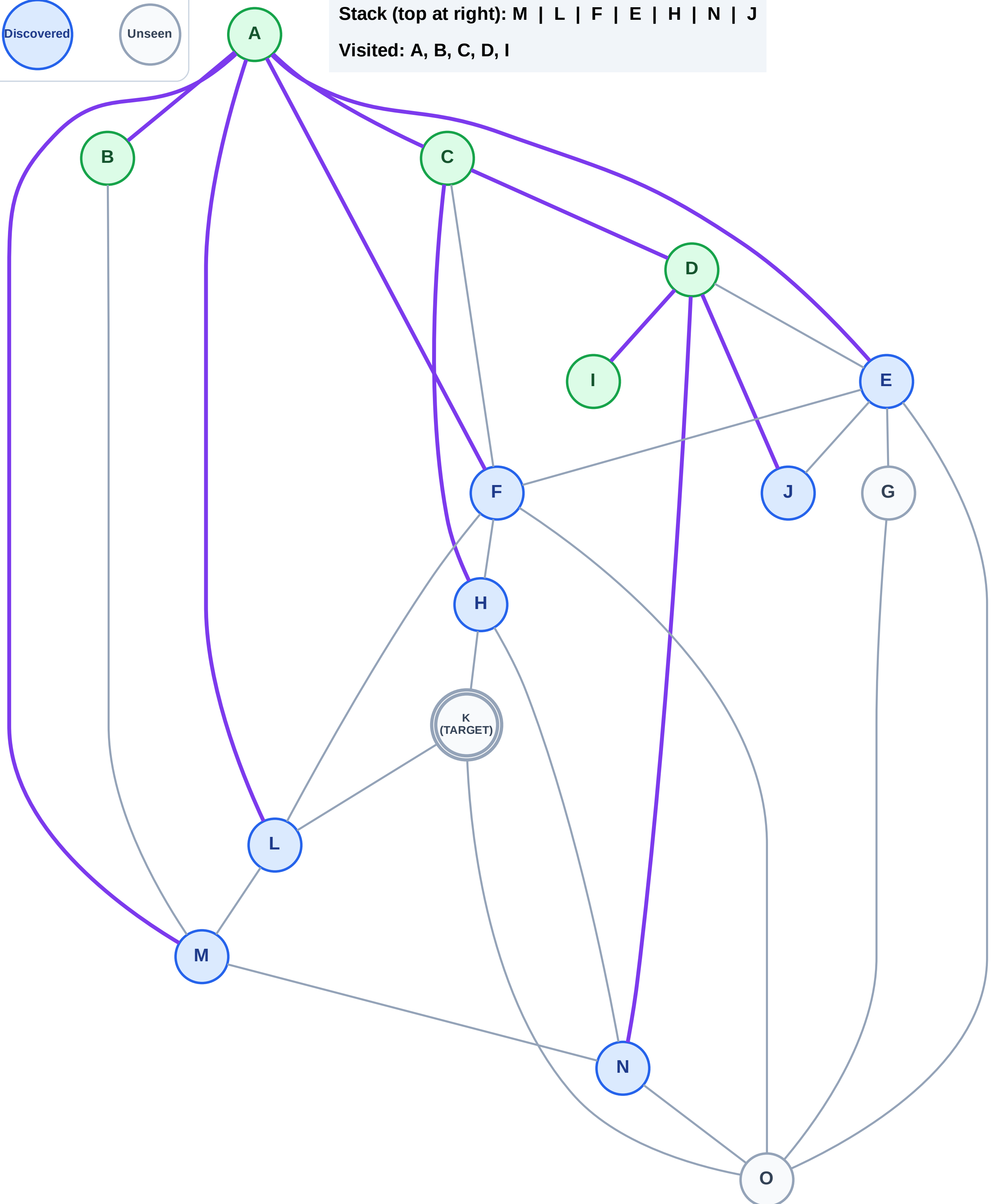
Visited: A, B, C, D, I

Complete



Discovered

Unseen



Step 12: Process node J

Legend

Complete

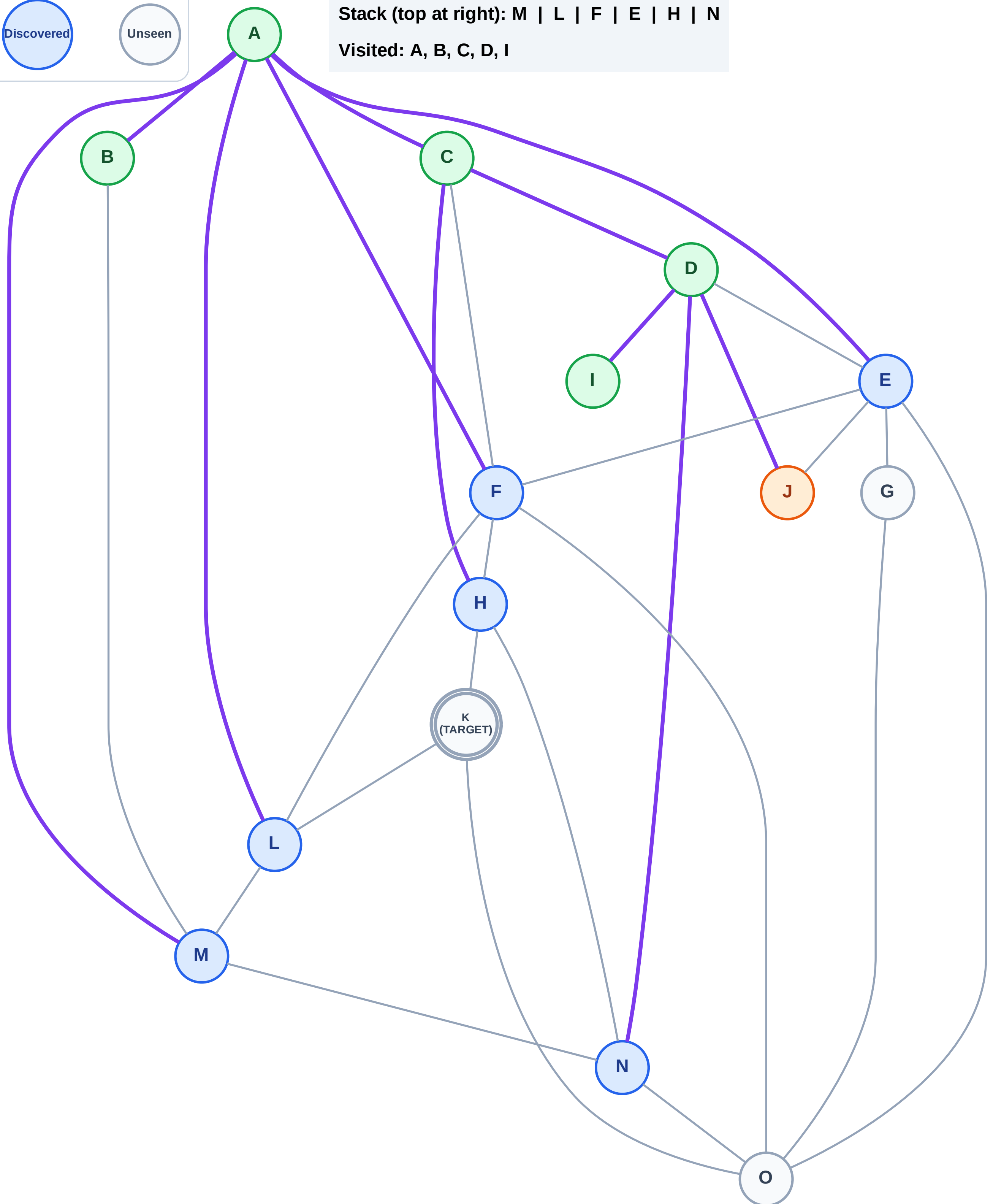
Processing

Discovered

Unseen

Stack (top at right): M | L | F | E | H | N

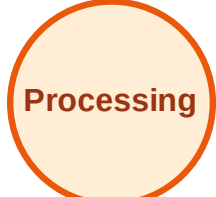
Visited: A, B, C, D, I



DFS Traversal

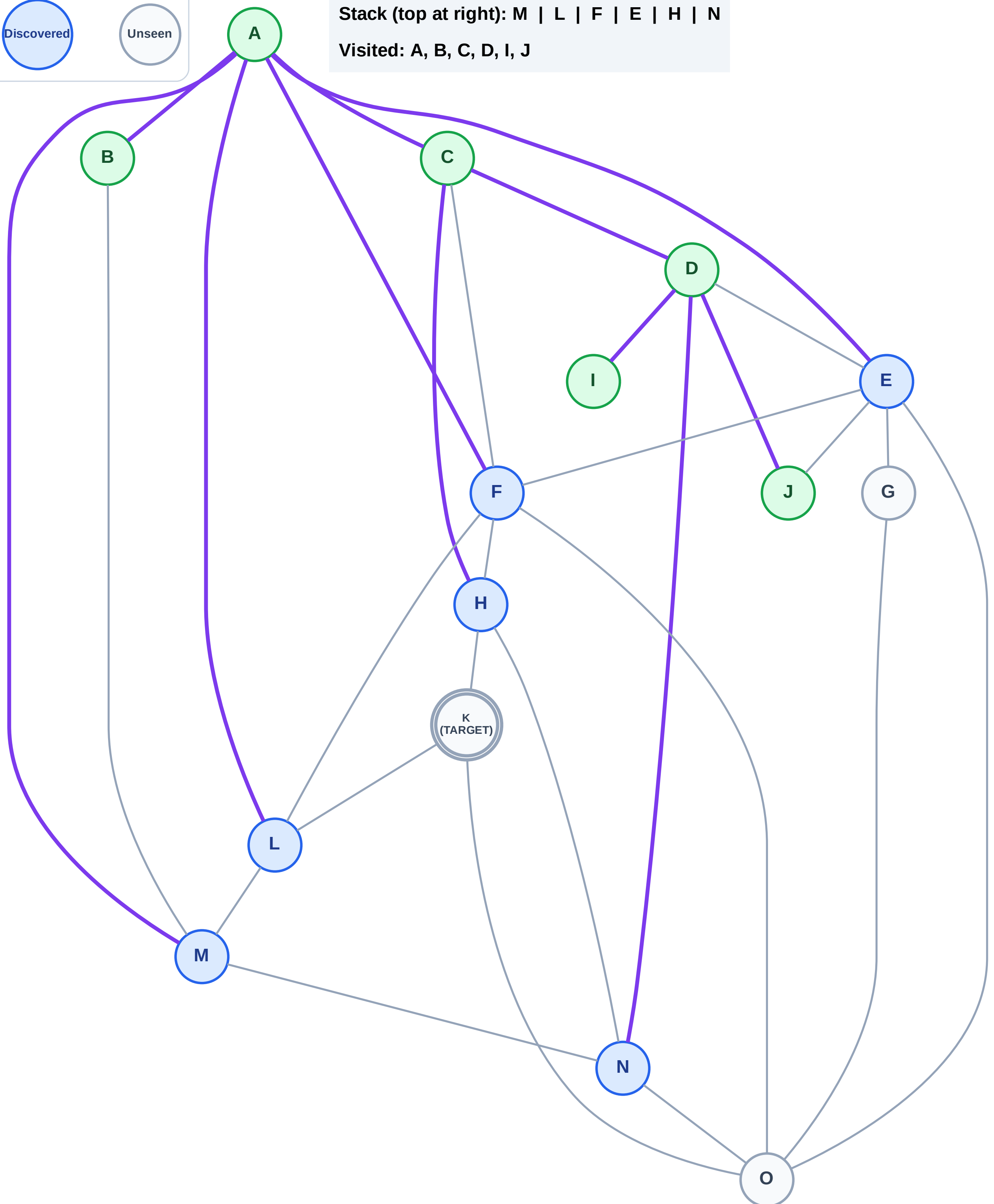
Step 13: No new neighbors from J

Legend



Stack (top at right): M | L | F | E | H | N

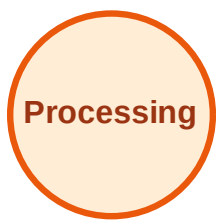
Visited: A, B, C, D, I, J



DFS Traversal

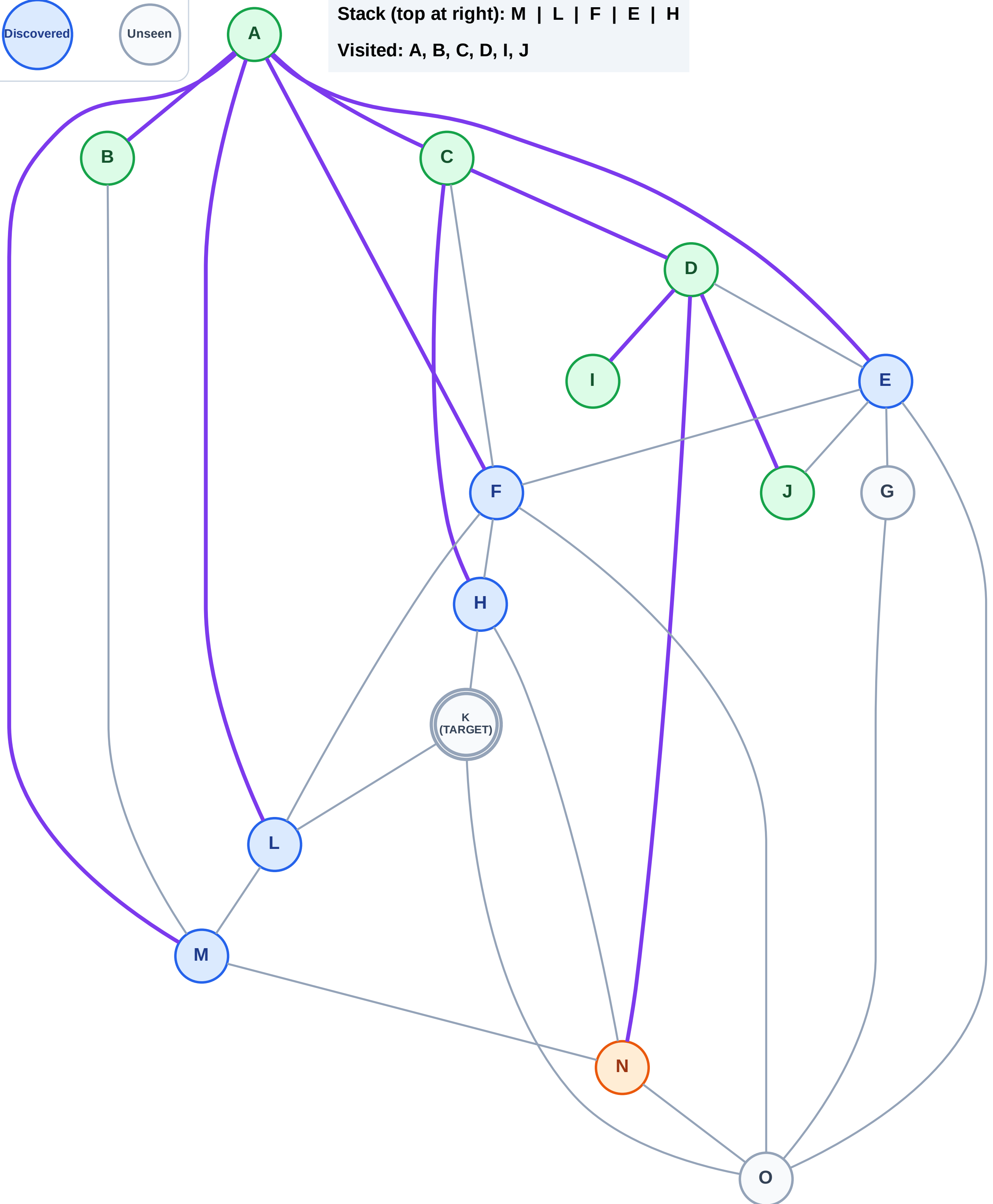
Step 14: Process node N

Legend



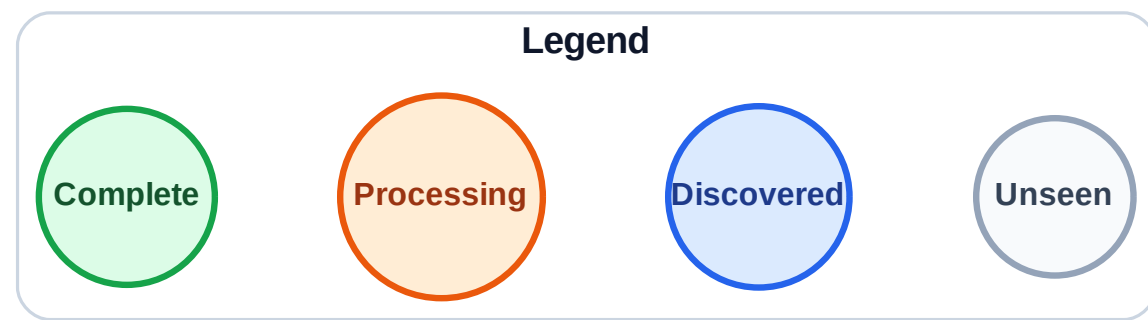
Stack (top at right): M | L | F | E | H

Visited: A, B, C, D, I, J



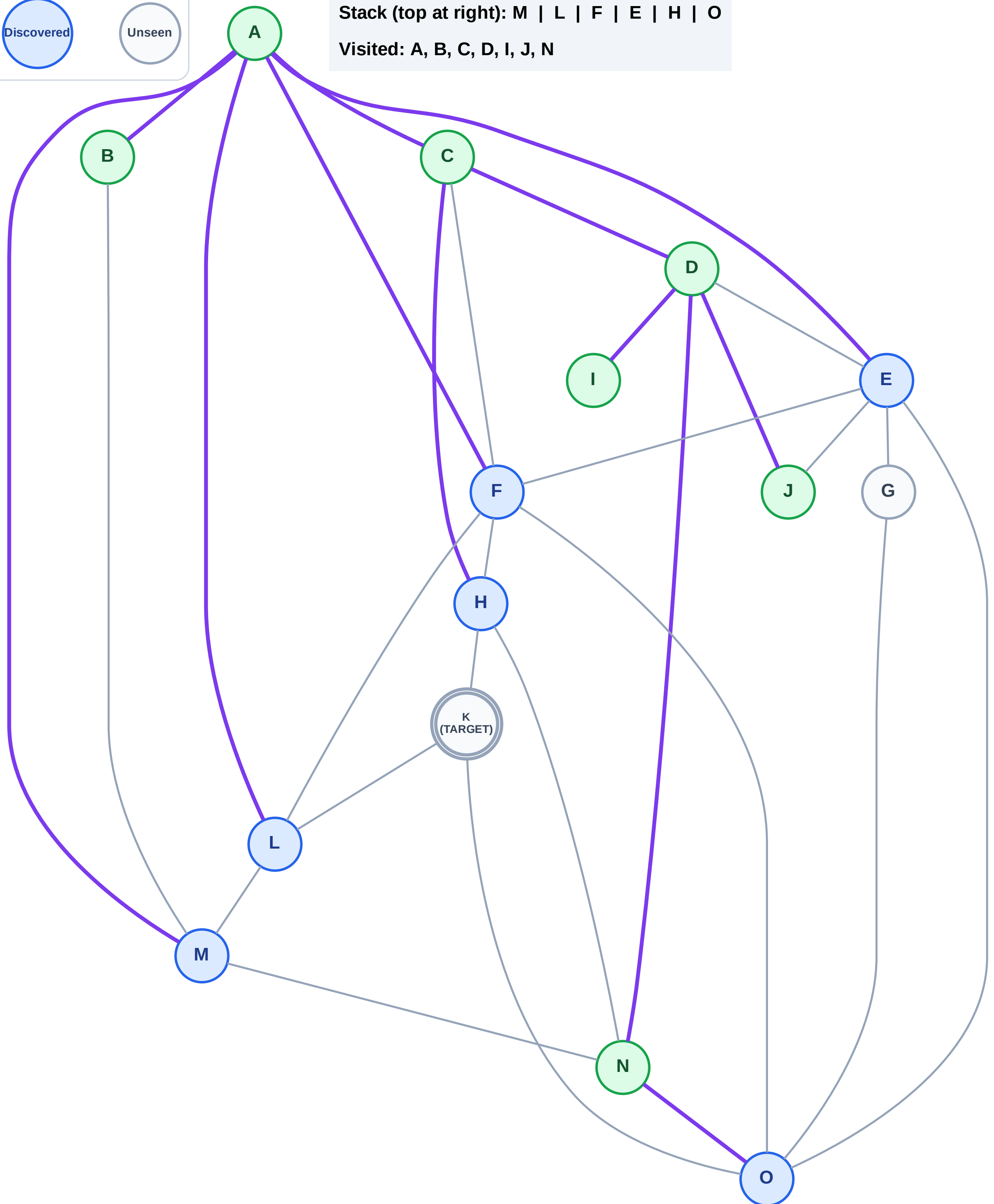
Step 15: Discover O from N

Step 15: Discover O from N



Stack (top at right): M | L | F | E | H | O

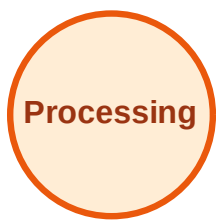
Visited: A, B, C, D, I, J, N



DFS Traversal

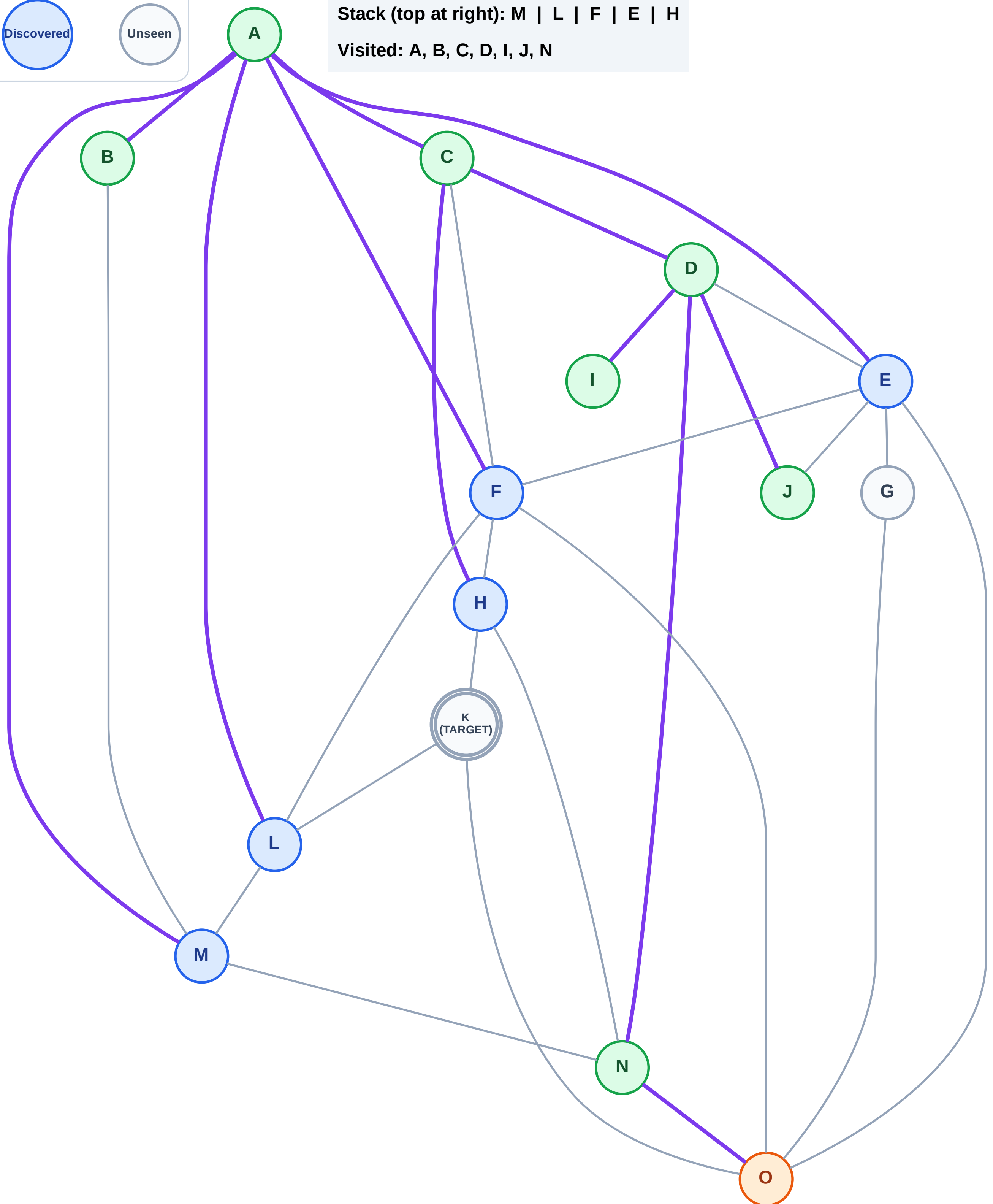
Step 16: Process node O

Legend



Stack (top at right): M | L | F | E | H

Visited: A, B, C, D, I, J, N



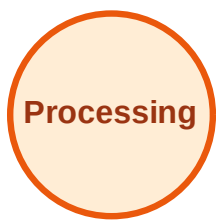
DFS Traversal

Step 17: Discover K, G from O

Legend



Complete



Processing



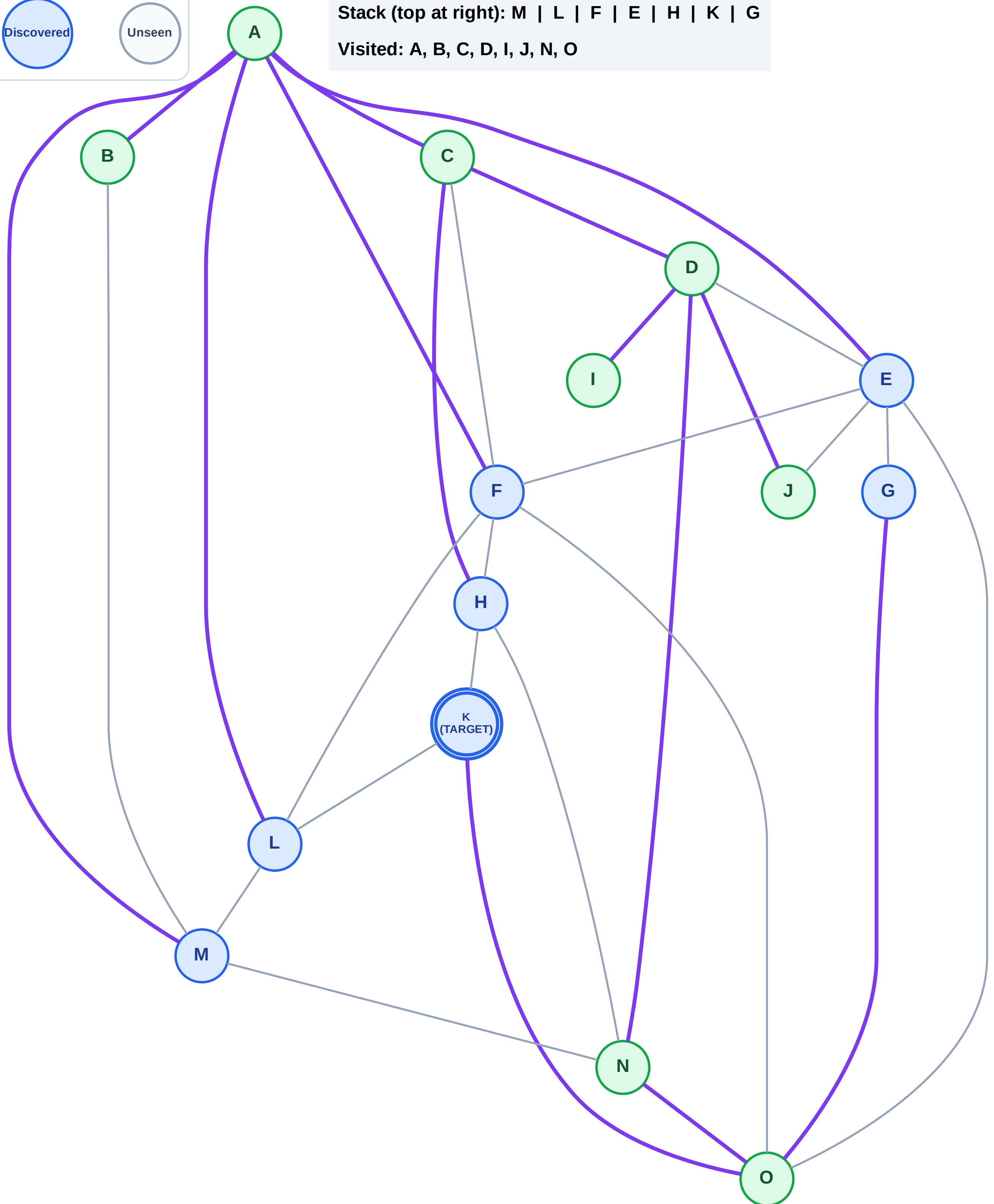
Discovered



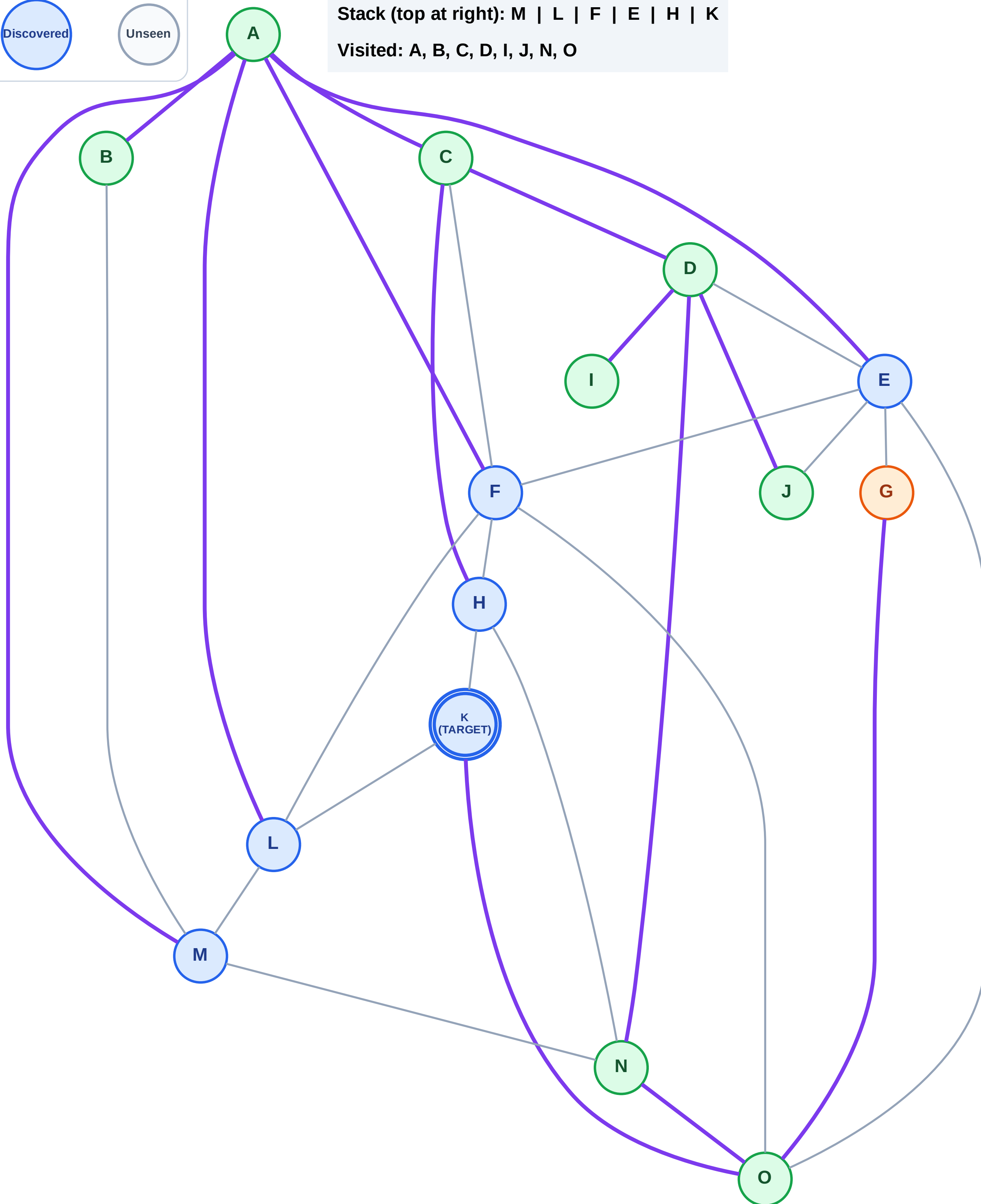
Unseen

Stack (top at right): M | L | F | E | H | K | G

Visited: A, B, C, D, I, J, N, O



Step 18: Process node G



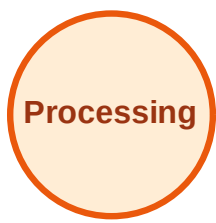
DFS Traversal

Step 19: No new neighbors from G

Legend



Complete



Processing



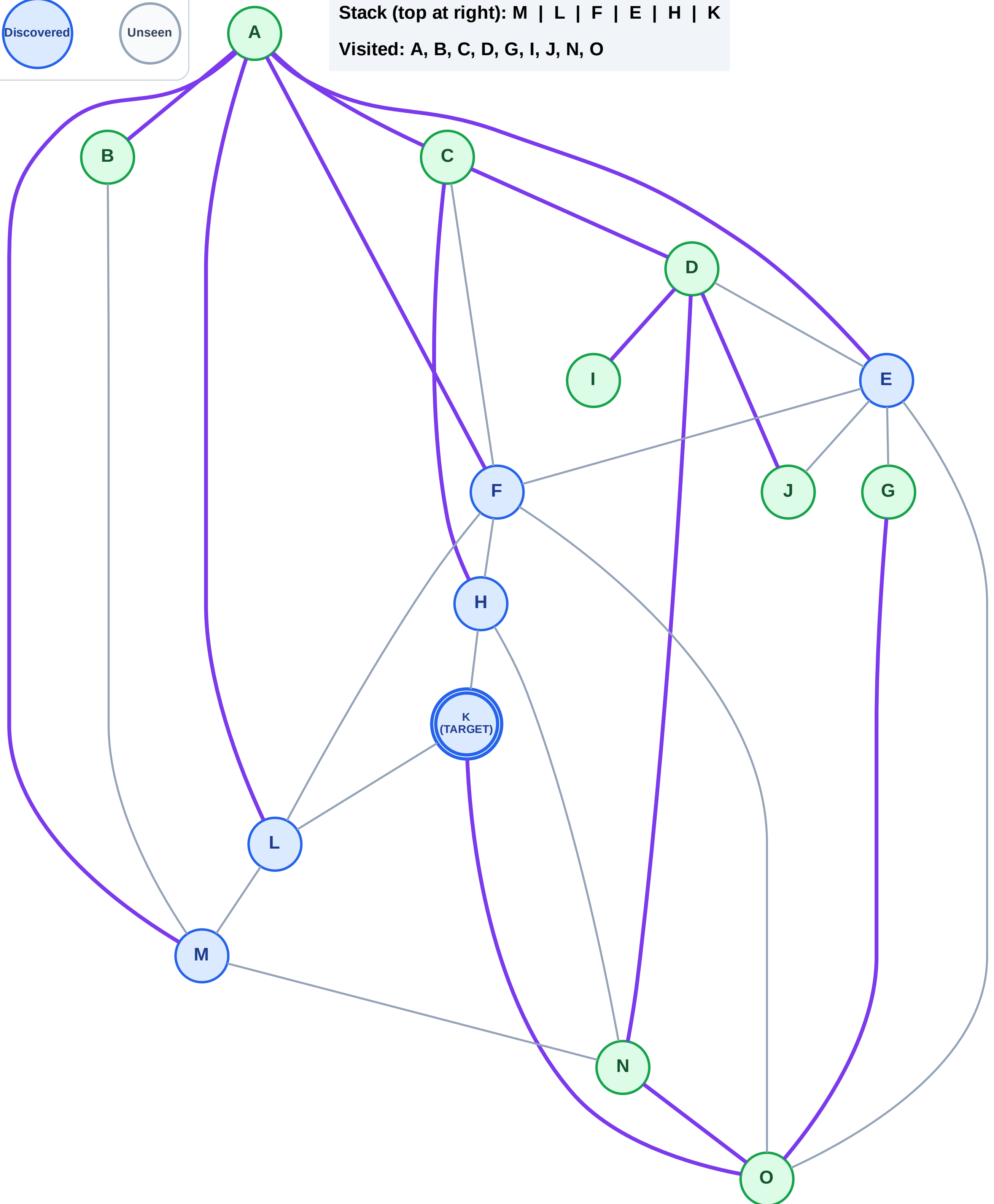
Discovered



Unseen

Stack (top at right): M | L | F | E | H | K

Visited: A, B, C, D, G, I, J, N, O



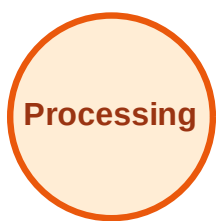
DFS Traversal

Step 20: Process node K

Legend



Complete



Processing



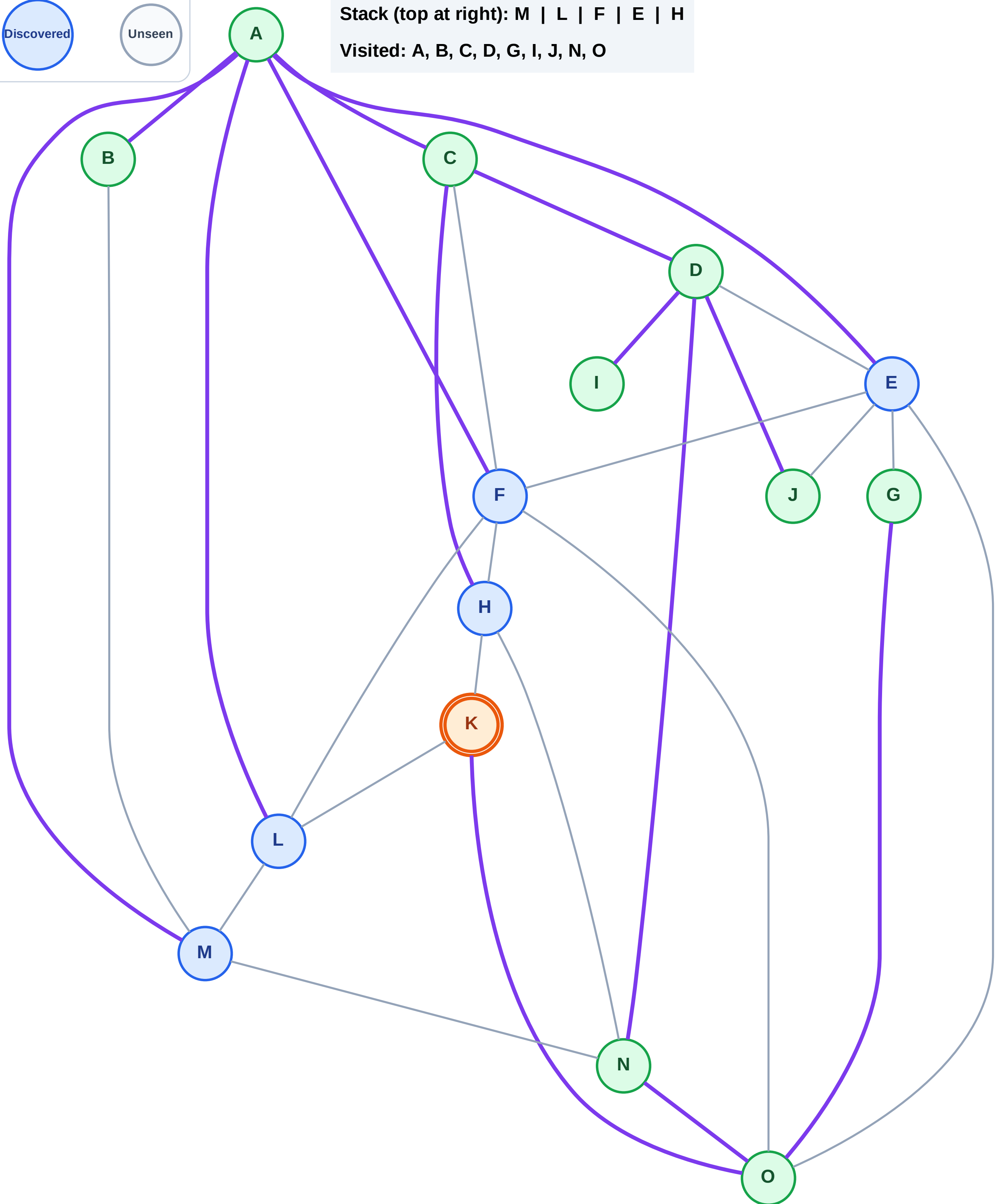
Discovered



Unseen

Stack (top at right): M | L | F | E | H

Visited: A, B, C, D, G, I, J, N, O



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (5 hops): A → C → D → N → O → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | F | E | H
Visited: A, B, C, D, G, I, J, K, N, O

